

Crowdsourced Forecasts and the Market Reaction to Earnings Announcement News*

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Abstract

This study examines whether crowdsourced forecasts of earnings and revenues help investors unravel bias in earnings announcement news, which is commonly derived from analyst forecasts. Our results suggest that investors, on average, understand and price the predictive signals reflected in crowdsourced forecasts about the bias in analyst-based earnings and revenue surprises. Using the staggered addition of firms to the Estimote platform, we find that crowdsourced coverage is associated with reductions in the mispricing of forecast bias and declines in earnings announcement premia. We further find some evidence that managers use income-increasing accruals to meet the crowdsourced forecast benchmark and that they respond to crowdsourced coverage through increased downward earnings and revenue guidance. Overall, we conclude that user-generated content on crowdsourced financial information platforms helps investors discount biases in traditional equity research and thereby better process the news in earnings announcements.

Keywords: Crowdsourcing, Estimote, sell-side analysts, forecast bias, earnings announcements, market reactions.

JEL Classifications: G14, G20, M41.

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1. Introduction

This paper examines whether user-generated content on crowdsourced financial information platforms helps investors unravel bias in earnings announcement news. Despite a pessimistic bias caused by the conflicts of interest in sell-side equity research, financial journalists and online media platforms frequently rely on consensus analyst forecasts when transmitting earnings announcement news to the broader market (see, e.g., [Call et al. 2022](#), and [Appendix A](#)). Given this fixation on analyst forecasts and investors’ reliance on the media around earnings announcements (e.g., [Lawrence et al. 2017](#); [Guest 2021](#)), it is not surprising that investors fail to fully undo the bias in analyst forecasts when measuring earnings news ([Ma and Markov 2017](#); [Veenman and Verwijmeren 2018](#); [Johnson et al. 2020](#)).

The recent rise of online financial analysis platforms, as part of the broader FinTech revolution, has the potential to crowd out traditional information intermediaries in capital markets and reduce the reliance on analyst forecasts. For example, [Drake et al. \(2023\)](#) find that financial analysis on SeekingAlpha is associated with attenuated market reactions to analyst research. In this paper, we focus on Estimote, an online platform that crowdsources forecasts of quarterly earnings and revenues from a wide range of individuals and aggregates these forecasts into a consensus estimate. [Jame et al. \(2016\)](#) find that, relative to analyst forecasts, the crowdsourced consensus is incrementally useful in forecasting earnings and in measuring the market’s expectations of earnings. Particularly pertinent to our study is the result that crowdsourced forecasts are substantially less biased than analyst forecasts, which suggests they provide a useful alternative benchmark to determine the news in an earnings announcement.

Because the bias that analysts incorporate in their forecasts is unobservable, investors cannot perfectly distinguish between positive “Street” earnings surprises that reflect unexpected good news and positive surprises that are caused by pessimistic bias in the consensus forecast. To assess the extent of forecast bias, investors should form priors about this bias based on observable signals, such as patterns in past earnings surprises and the forecast

trajectory over the quarter. Given these priors, we expect investors to discount positive earnings surprises when they believe analysts have understated their forecasts (Chen et al. 2005; Abarbanell and Park 2017).

The extent to which investors discount bias in earnings and revenue surprises depends on their ability to obtain and process signals that help them to form expectations about forecast bias (Bloomfield 2002; Blankespoor et al. 2020). When investors’ priors based on these signals are relatively more precise, they should place more weight on a bias-adjusted measure of earnings surprise when reacting to earnings announcements, and less weight on biased Street earnings surprises. Prior evidence on the mispricing of earnings surprise bias suggests that investors face non-trivial costs in processing such signals and that they place too much weight on biased Street earnings surprises when interpreting earnings announcements (Ma and Markov 2017; Veenman and Verwijmeren 2018).

The information processing costs of forecast-bias signals should decline when these signals become more visible to investors. We argue that compared to widely used platforms such as Yahoo! Finance, Bloomberg, Factset, or SeekingAlpha, which display earnings and revenue surprises based on analyst forecasts, Estimote can enhance the visibility of forecast bias and expose this bias. The reason is that Estimote displays histories of quarterly earnings and revenues alongside the Street *and* alternative, less-biased, consensus forecasts. Accordingly, we contend that alternative forecasts generated by online contributors, who are not subject to the conflicts of interest in sell-side equity research, can improve investors’ assessments of earnings news by exposing the bias in traditional analyst forecasts.

We construct a sample of quarterly earnings announcements of firms covered by analysts and Estimote contributors during 2012–2020. Our initial descriptive analyses provide the following insights. First, we confirm the incremental usefulness of crowdsourced earnings forecasts in measuring market earnings expectations (Jame et al. 2016). Second, we find that crowdsourced forecasts of *revenues* are relevant in explaining earnings announcement returns, incremental to the information in analyst-based revenue surprises (e.g., Hand et al.

2021). Third, we illustrate that the pessimistic bias and walkdown in analyst earnings and revenue forecasts (Richardson et al. 2004) are substantially weaker for crowdsourced estimates. Finally, consistent with crowdsourced forecasts providing a relevant alternative benchmark for earnings announcement news, we find that the return premia for meeting analyst earnings and revenue expectations are largely eliminated when firms fail to meet crowdsourced expectations.¹

Given these descriptive insights, we examine whether crowdsourced forecasts help investors form priors about the expected bias in analyst forecasts. We first examine whether the pre-earnings announcement difference between crowdsourced and analyst forecasts predicts the bias in Street earnings and revenue surprises and whether this predictive value is priced before earnings announcements. The idea of this test is that if investors understand and price the signals about expected bias in earnings and revenue surprises, they should not be surprised by this component of “news.” Consistent with this idea, we find that the forecast difference has substantial predictive power for the sign of the earnings or revenue surprise, but it is not associated with stock returns around earnings announcements.

We next examine the implications of crowdsourced coverage for investors’ assessments of forecast bias. For these tests, we follow Jame et al. (2022) by exploiting the staggered addition of firms to the Estimote platform and expanding our sample to include the earnings announcements of firms before they are covered by Estimote and other firms never included on Estimote during our sample period. Using ex-ante measures of expected forecast bias (Veenman and Verwijmeren 2018, 2022), we find that investors misprice the predictable bias in earnings and revenue surprises before Estimote coverage and for firms never covered by Estimote. After firms are added to the Estimote platform, however, we find significant reductions in this mispricing. For earnings surprises, we no longer find evidence of mispricing after crowdsourced coverage.

Given investors’ improved information processing, we examine the effects of crowdsourced

¹Unless we explicitly refer to firms *exactly* meeting or *just* beating expectations, we use the term “meet(ing) expectations” to refer to firms meeting or beating expectations throughout the paper.

coverage on firms’ average earnings announcement returns. Prior research finds that firms, on average, earn positive abnormal returns around earnings announcements (e.g., [Cohen et al. 2007](#)). [Johnson et al. \(2020\)](#) link these earnings announcement premia to investors’ mispricing of predictable bias in earnings surprises. Given the reduced mispricing of forecast bias, we expect that earnings announcement premia decline following crowdsourced coverage.

A second channel through which crowdsourced coverage can lower average earnings announcement returns is through a reduction in analyst forecast pessimism ([Jame et al. 2022](#)), which leads to lower earnings and revenue surprises. Although these two channels are not mutually exclusive and hard to separate, we can control for changes in the levels of earnings and revenue surprises when examining the effects of crowdsourced coverage on earnings announcement returns. Extending the results of [Jame et al. \(2022\)](#) on reduced analyst forecast bias, we first demonstrate that crowdsourced coverage is associated with reductions in the likelihood that firms marginally beat earnings expectations, meet analyst revenue expectations, or simultaneously meet earnings and revenue expectations. Next, we find that relative to non-covered firm-quarters, crowdsourced coverage results in around 50 basis points lower average announcement returns. This result does not change materially when we control for earnings and revenue surprises or when we condition on the sign of the surprise, which suggests that the lower announcement returns are incremental to the effects of the second, reduced-forecast-bias, channel.

Lastly, we study whether the additional performance benchmark, and the associated reductions in forecast bias and capital market benefits, affect managers’ use of accrual management and guidance to meet expectations. Using a one-step model to isolate quarterly “abnormal” working capital accruals ([Chen et al. 2018](#)), we find evidence of (i) higher abnormal accruals when firms meet both analyst and crowdsourced earnings expectations compared to when they meet analyst expectations only, and (ii) higher abnormal accruals in quarters after firms miss crowdsourced expectations. However, we find no evidence of changes in the use of accruals to meet analyst expectations after firms are added to Estimate. Instead, we

find significant increases in the likelihood that firms provide downward earnings or revenue guidance.²

Collectively, we conclude that the alternative, less biased, forecasts of earnings and revenues provided by crowdsourced coverage are associated with improvements in investors' assessments of analyst forecast bias and reduced earnings announcement premia. Although we find some evidence that managers increase their use of public guidance to lower expectations after crowdsourced coverage, our results suggest that this response is not sufficient to counter the reductions in earnings announcement returns and forecast bias after Estimote coverage. At the same time, we stress that despite the improvements in information processing and the reductions in forecast bias we find, the pessimistic bias in analyst earnings and revenue forecasts remains an important issue as the majority of firms still meet these biased expectations in most quarters.

This paper contributes to the nascent literature on the importance of social media, the internet, and online user-generated content for the pricing of accounting information in capital markets (e.g., [Blankespoor et al. 2014](#); [Chen et al. 2014](#); [Jame et al. 2016](#); [Drake et al. 2017](#); [Tang 2018](#); [Campbell et al. 2019](#); [Farrell et al. 2022](#); [Gomez et al. 2022](#)) and its disciplining effects on analysts and other market participants (e.g., [Dube and Zhu 2021](#); [Jame et al. 2022](#)). We find that crowdsourced financial research helps expose biases in traditional forms of equity research and improves investors' ability to see through these biases around earnings announcements. Our results illustrate how online crowdsourced financial research can alter the dynamics and consequences of the strategic interaction between managers and

²Although not the primary focus of our study, it is important to note that these tests relate to concurrent work that also examines the effects of Estimote coverage on managers. [Sul \(2020\)](#) similarly finds an increase in downward earnings guidance for firms after Estimote coverage. Using annual two-step estimates of real earnings management, he further concludes that Estimote coverage is associated with increases in real activities manipulation. [Ott et al. \(2020\)](#) draw similar conclusions regarding real activities manipulation and accrual management using two-step estimates and annual accounting variables. Using one-step estimates of quarterly abnormal accruals that address measurement concerns with the use of residuals as dependent variable ([Chen et al. 2018](#)), we do not find that coverage on Estimote is associated with an increased use of accrual-based earnings management to meet earnings expectations. Also, note that it is conceptually less clear how the potential pressure to meet *quarterly* earnings expectations maps into the management of *annual* earnings as studied in [Sul \(2020\)](#) and [Ott et al. \(2020\)](#).

analysts. In particular, our study helps explain why recent studies find that crowdsourced research changes the behavior and added value of incumbent market participants, such as sell-side analysts (Banker et al. 2017; Jame et al. 2022; Drake et al. 2023).³

We also contribute to the literature on the pricing of earnings announcement news. Recent research examines the drivers of market reactions to earnings announcements beyond the traditional analyst-based surprise in Street earnings (Bradshaw et al. 2018; Givoly et al. 2019; Beaver et al. 2020; Hand et al. 2021). Our study illustrates the importance of online user-generated content in providing an alternative benchmark to derive the news in reported earnings and revenues. We further find that these forecasts can change the positive valuation effects from reporting positively biased earnings surprises (Bartov et al. 2002; Bhojraj et al. 2009) and reduce the mispricing of predictable variation in earnings surprises (Ma and Markov 2017; Veenman and Verwijmeren 2018).

2. Background, institutional setting, and prediction

2.1. *Sell-side analyst forecasts and market earnings expectations*

Financial information platforms such as Yahoo! Finance, Bloomberg, and Factset rely on sell-side analyst forecasts when depicting firms’ performance expectations and measures of earnings and revenue surprise (see Appendix A). The financial press similarly uses analyst forecasts as input to determine the news content of earnings announcements. Call et al. (2022) find that over 70 percent of their surveyed financial journalists are likely to compare earnings to analyst expectations when reporting on an earnings announcement. Anecdotal examples from the media further confirm the use of analyst forecasts in determining the direction of earnings announcement news.⁴ Given the importance of these platforms and the

³Our research also relates to prior work on “whisper” forecasts. Bagnoli et al. (1999) find that whisper forecasts incrementally capture market earnings expectations. They also find that whisper forecasts contain an optimistic bias instead of the pessimistic bias in analyst forecasts but do not examine the consequences of this bias for the market reaction to earnings announcements. Unlike crowdsourced forecasts, the processes used to create whisper forecasts are opaque and there is mixed evidence of their value relevance in recent years see (Jame et al. 2016).

⁴See, e.g., <https://www.cnbc.com/2022/02/01/alphabet-google-q4-2021-earnings.html> and evidence from algorithmically-generated “robo-journalism” articles which typically rely on the analyst consensus in deter-

media as sources of information around earnings announcements (e.g., [Bushee et al. 2010](#); [Drake et al. 2014](#); [Lawrence et al. 2017](#); [Guest 2021](#)), this reliance on analyst forecasts is likely to influence investors’ reactions to earnings news.

An important concern with the reliance on analyst forecasts around earnings announcements is that conflicts of interest in sell-side equity research cause these forecasts to be pessimistically biased (e.g., [Richardson et al. 2004](#)).⁵ Prior research finds this pessimistic bias is related to managers’ incentives to avoid negative surprises and analysts’ incentives to help firms achieve this objective. For example, forecast bias can help analysts gain private access to management ([Das et al. 1998](#); [Ke and Yu 2006](#); [Hilary and Hsu 2013](#)), investment banking businesses for their employers ([Feng and McVay 2010](#)), and other benefits such as the ability to sponsor non-deal roadshows ([Bradley et al. 2022](#)).⁶ Because of the pessimistic bias in short-term forecasts, measures of news derived from analyst-based earnings surprises are positively biased.

Prior studies find that investors learn about analyst forecast attributes, such as accuracy and bias. For example, [Chen et al. \(2005\)](#) find that investors place more weight on the news in an analyst’s forecast when the analyst has accumulated a longer track record of forecasting performance. Other studies find that investors partly discount the bias in earnings news derived from analyst forecasts (e.g., [Keung et al. 2010](#); [Abarbanell and Park 2017](#)). [Abarbanell and Park \(2017\)](#) present a framework in which investors anticipate the bias in an earnings surprise and discount announcements of firms with a higher ex-ante probability of reporting a positively biased surprise. Their empirical results suggest that investors partition

mining the news content in earnings announcements ([Blankespoor et al. 2018, 2019](#)).

⁵This pessimistic bias appears at odds with the well-known tendency for analysts to be overly optimistic about the firms they cover, for example, due to incentives to secure investment banking deals or to generate trading commissions (e.g., [Cowen et al. 2006](#); [Lin and McNichols 1998](#)). While such incentives cause analysts’ long-term forecasts, stock recommendations, and price targets to be optimistically biased, the short-term forecasts we focus on induce incentives for pessimistic bias because they determine the direction of earnings announcement news. See [Veenman and Verwijmeren \(2018\)](#) for further discussion on the difference between bias in short- versus long-term forecasts.

⁶Although Regulation Fair Disclosure prohibits firms’ selective disclosure of material information to individuals, empirical evidence suggests that access to management remains a key source of information for analysts ([Soltes 2014](#); [Green et al. 2014](#); [Brown et al. 2015](#)).

the earnings surprise into a true news component and an expected bias component.

However, recent work also finds that investors’ learning is incomplete and investors fail to *fully* undo the bias in analyst-based earnings surprises. [Ma and Markov \(2017\)](#) model firms’ propensity to meet expectations and find that a trading strategy based on the market’s prediction errors yields significant abnormal returns. [Veenman and Verwijmeren \(2018\)](#) document that investors fail to fully unravel earnings surprise bias based on information about analysts’ past forecast errors. [Johnson et al. \(2020\)](#) find that earnings surprise bias induces predictable reversals in earnings announcement returns and explains the earnings announcement premium.⁷ Collectively, these insights suggest that investors, on average, misprice the predictable component of analyst forecast bias. [Ma and Markov \(2017\)](#) and [Veenman and Verwijmeren \(2018\)](#) conclude that this mispricing is consistent with investors facing constraints in using public signals to ex-ante assess the probability that firms meet expectations. This interpretation comports with models of costly information processing, in which prices do not fully reflect public information because investors face constraints in processing information (e.g., [Blankespoor et al. 2020](#)).

2.2. Crowdsourced financial research

Online financial analysis platforms have started to disrupt the traditional business models in the finance industry. Prominent examples include SeekingAlpha, a crowdsourced financial investment research platform, and Estimize, a crowdsourced financial estimates platform.⁸ We focus on crowdsourced forecasts on Estimize, which aims to provide a measure of market expectations that is different from the analyst consensus. Estimize has a significant

⁷[Berger et al. \(2019\)](#) further find that analysts’ selectivity in revising their short-term forecasts in response to positive information creates more beatable forecasts and is associated with predictable earnings announcement returns. Their results suggest investors fail to fully see through this strategic behavior.

⁸The term crowdsourcing originates from the words *crowd* and *(out)source* and describes the activity of transferring tasks to the large public to create content, products, or ideas ([Howe 2006](#); [Schenk and Guittard 2011](#)). According to the theory of the wisdom of crowds ([Surowiecki 2004](#)), the collective intelligence obtained from the aggregation of private information from a large group of individuals can generate smarter solutions to a problem than the information provided by a single expert. The success of crowdsourcing depends on members’ diversity, independence, and their contribution of local, decentralized knowledge.

contributor base and its forecasts are available to investors through a variety of prominent sources (e.g., Bloomberg). Contributors include non-financial professionals, such as students and academics, and financial professionals, such as sell-side, buy-side, and independent analysts.⁹ In line with prior research on individuals’ motivations to participate in crowdsourcing initiatives more generally, the platform is set up to incentivize contributors to improve their reputation, visibility, and forecasting skills.¹⁰ Estimote performs various quality checks to assure forecast accuracy and mitigate manipulation. It verifies the quality of submitted forecasts and eliminates extreme and suspect forecasts from the consensus, benchmarks contributors’ forecast accuracy, and manually reviews estimates from new contributors.

Given the rise of these platforms, recent research has started to study their role and significance in capital markets. Empirical evidence suggests these new sources of investment research provide relevant incremental information for forecasting and valuation (e.g., [Chen et al. 2014](#); [Jame et al. 2016](#); [Farrell et al. 2022](#); [Drake et al. 2023](#)). For example, [Jame et al. \(2016\)](#) find that, relative to analyst forecasts, crowdsourced forecasts are incrementally useful in forecasting earnings and in measuring the market’s expectations of earnings. They also find that crowdsourced earnings forecasts are substantially less pessimistic, which suggests that investors can use this alternative benchmark to determine the news content of an earnings announcement.¹¹ [Jame et al. \(2022\)](#) conclude that crowdsourced research disciplines analysts, as their forecasts become less biased after Estimote initiates coverage of a firm. [Banker et al. \(2017\)](#) further find that analysts become more responsive to earnings announcements after Estimote initiates coverage of a firm.¹²

⁹Although contributors are not required to disclose their names and professional backgrounds, most *active* contributors make their professional backgrounds publicly available. In this regard, [Brown and Khavis \(2018\)](#) conclude that the biographies of Estimote contributors are reliable.

¹⁰See e.g., [Wasko and Faraj \(2005\)](#) and [Brabham \(2010\)](#). [Crawford et al. \(2018\)](#) support these insights and find that buy-side analysts share their knowledge freely on online platforms to improve their reputation and career outcomes and assure that prices reflect fundamental value.

¹¹Consistent with the wisdom of crowds effect, the accuracy and value of crowdsourced forecasts increase with the number of contributors ([Jame et al. 2016](#)) and their diversity and decentralization ([Adebambo et al. 2016](#)).

¹²[Da and Huang \(2020\)](#) demonstrate the limits to crowdsourced research and find that herding causes contributors to put less weight on their private information, which reduces the accuracy of consensus estimates. To increase the extent of private information incorporated in crowdsourced forecasts, Estimote

2.3. Alternative forecasts and the pricing of predictable forecast bias

We argue that alternative forecast benchmarks that are not affected by the conflicts of interest in sell-side equity research—such as crowdsourced forecasts on Estimize—can expose bias in analyst forecasts and facilitate investors’ processing of earnings announcement news. The argument that crowdsourced forecasts expose bias in analyst forecasts follows [Jame et al. \(2022\)](#), who focus on the effects of crowdsourced coverage on incumbent analysts and find that competition from crowdsourced research is associated with reductions in analyst forecast pessimism. We extend this argument and examine investors’ valuations of earnings announcement news and the unraveling of predictable forecast bias.

We expect the crowdsourced forecast benchmark to influence investors’ unraveling of analyst forecast bias through a reduction in information processing costs ([Blankespoor et al. 2020](#)). Because analysts’ private information is unknown, the bias they incorporate in their forecasts is unobservable to outsiders. Consequently, investors cannot perfectly distinguish between positive earnings surprises that reflect unexpected good news and those caused by pessimistic forecasts. To assess the extent of forecast bias, investors should form priors about this bias based on observable signals, such as firms’ earnings surprise history. Given their priors, investors are expected to discount positive earnings surprises when a firm’s earnings surprise history suggests analysts are likely to be pessimistic.

The extent to which investors discount biased earnings and revenue surprises depends on their ability to acquire and process information signals that help them to form expectations about forecast bias ([Bloomfield 2002](#); [Blankespoor et al. 2020](#)). When investors form relatively more precise (imprecise) priors about expected forecast bias, we expect them to place more (less) weight on a bias-adjusted measure of earnings surprise when evaluating earnings announcements, and less (more) weight on biased Street earnings surprises (e.g., [Chen et al. 2005](#)). Prior research, however, suggests that investors face non-trivial information processing costs and place too much weight on biased earnings surprises when interpreting earnings

disabled contributors’ view of the Estimize consensus as of November 2015.

news (Ma and Markov 2017; Veenman and Verwijmeren 2018).

The costs to process predictive signals of forecast bias should decline when these signals become more visible. Compared to widely used financial information platforms, which display earnings and revenue surprises based on analyst forecasts, the Estimote platform can enhance the visibility of forecast bias and expose this bias because it displays histories of earnings and revenues alongside the Street *and* alternative, less-biased, consensus forecasts. Figure 1 illustrates this point and shows that Netflix reported earnings that met analyst expectations in seven out of eight quarters, but missed crowdsourced expectations in four of these quarters. This example demonstrates how the use of an alternative forecast benchmark has important consequences for the direction of earnings news that is implied by the forecasts, and how it can be used to assess analyst forecast bias beyond the signals in firms’ past earnings surprises. By observing the ex-ante difference between analyst and crowdsourced forecasts, we expect that investors form more precise priors about the probability that a firm will report earnings or revenues that meet analyst expectations.

Overall, we predict that user-generated content on online financial information platforms exposes the bias in analyst forecasts and improves investors’ assessments of earnings announcement news. At the same time, given prior evidence of mispricing and the widespread reliance on the analyst consensus by financial information platforms and the media—as illustrated in Appendix A for Apple—we argue that it is ex-ante unclear whether investors place sufficient weights on alternative forecast benchmarks.

3. Data and descriptive statistics

3.1. Sample selection and data sources

Panel A of Table 1 presents the sample selection details. We first construct a sample of firm-quarters during 2012–2020 in the intersection of CRSP, Compustat, and I/B/E/S. We obtain forecasts and actuals of quarterly EPS and revenues from the I/B/E/S unadjusted detail files and measure the sell-side (mean) consensus forecast using each analyst’s most

recent forecast in the 120 days before the announcement.¹³ We then measure Street earnings and revenue surprises (ES_{Street} and SS_{Street}) as the difference between I/B/E/S actuals and the consensus forecasts.¹⁴ To ensure we accurately measure earnings announcement dates, we (i) eliminate observations for which I/B/E/S and Compustat disagree about the announcement date by more than two trading days (Cready et al. 2014; Johnson and So 2018), (ii) adjust the date to the earlier of the Compustat and I/B/E/S dates (Dellavigna and Pollet 2009), and (iii) adjust the date to the next trading day when the announcement is made after trading hours (Johnson and So 2018). We also eliminate late announcements that occur more than 120 days after the fiscal quarter-end.

We merge this initial sample with firm-quarters on Estimize. Similar to the analyst consensus forecasts, we construct the crowdsourced consensus based on contributors’ latest forecasts in the 120 days before the earnings announcement. Following Jame et al. (2016), we exclude forecasts that Estimize flags as unreliable. Based on this consensus forecast, the variable ES_{Crowd} captures the crowdsourced earnings surprise. Variable SS_{Crowd} captures the crowdsourced revenue surprise. After merging the Estimize data with our initial sample, we obtain a sample of 48,586 firm-quarters of 2,668 firms. We use this Estimize sample in Tables 2 and 3 to examine how investors price the information in crowdsourced forecasts relative to analyst forecasts.

In other tests, we exploit the staggered addition of firms to the Estimize platform, which allows for stronger causal inference regarding the effects of crowdsourced coverage. Because this design requires a control group and pre-coverage period, we create an alternative sample that extends back to 2010 and does not require Estimize data. This sample contains firms never covered by Estimize during our sample period and firms added to Estimize in any

¹³Results are not sensitive to measuring the consensus forecast using shorter windows. The choice to include forecasts up to 120 days before the earnings announcement is consistent with Jame et al. (2022) and motivated by the observation that most analysts issue or update their forecasts shortly after the previous-quarter earnings announcement (e.g., Clement et al. 2011, Figure 1).

¹⁴Following Diether et al. (2002) and Payne and Thomas (2003), we use unadjusted forecasts and actuals of earnings, which we adjust for stock splits between the forecast and earnings announcement dates using split-adjustment factors from CRSP.

quarter during the years 2012–2016. We focus on additions during 2012–2016, because the quarterly addition of firms drops significantly after 2016. We further require firms to have at least four quarters in I/B/E/S before and after Estimize coverage, to ensure additions are not driven by IPOs.¹⁵ The extended sample consists of 128,182 firm-quarters for 5,456 firms. Additional data requirements reduce the sample size for some of our tests.

3.2. Descriptive statistics

Panel B of Table 1 presents descriptive statistics on the composition of the Estimize sample and forecast characteristics over time. The number of firm-quarters ranges from 1,614 in 2012 to a high of 7,458 in 2017. While Estimize covered about 12.8 percent of I/B/E/S firm-quarters in 2012, this proportion grew to 56.7 percent in 2017. The average CRSP size deciles suggest that firm size declines over the sample years. This decline reflects the addition by Estimize of relatively smaller firms in later years. Still, the average sample firm is much larger than the average listed firm (mean size decile of 8.1).

Focusing on the number of EPS forecasts in the crowdsourced and sell-side consensus, we find that the median firm-quarter in our sample receives more sell-side coverage (nine analysts) than crowdsourced coverage (six contributors). We also find a decline in analyst coverage over the sample period that reflects the initiation of Estimize coverage of smaller firms. Consistent with [Jame et al. \(2016\)](#), crowdsourced forecasts are issued much closer to the earnings announcement (median age around 18 days) than sell-side forecasts (median age around 68 days). This difference in forecast age persists over the sample period, although crowdsourced forecasts are submitted somewhat earlier in time in recent years.

Panel C of Table 1 reveals that the median firm-quarter’s earnings exceed analyst expectations by three cents per share (ES_{Street}), consistent with a pessimistic bias.¹⁶ The median

¹⁵In our sample, the number of firms added in 2012–2016 equals 802, 369, 353, 342, and 323, respectively. As of 2017, this number drops to 49 and below.

¹⁶All variables are defined in Appendix B. For presentation purposes, we winsorize continuous variables at the 1st and 99th percentiles of their distributions to account for the potential influence of extreme values on the means and standard deviations. We do not winsorize the announcement return variable because some of our tests focus on the prediction of announcement returns to identify mispricing. Winsorizing the dependent

crowdsourced surprise of one cent (ES_{Crowd}) suggests that the crowdsourced consensus is less pessimistic. Firms also meet crowdsourced expectations much less frequently (62 percent of firm-quarters) than analyst expectations (74 percent). We similarly find a pessimistic bias in analyst revenue forecasts that is stronger than for crowdsourced forecasts: firms meet analyst revenue expectations in 61 percent of firm-quarters, compared to 53 percent for crowdsourced revenue forecasts. Despite the reductions in forecast bias caused by crowdsourced coverage (Jame et al. 2022), the high frequencies of meeting analyst expectations for our sample firms suggest that Street earnings and revenue surprises remain substantially biased.

Figure 2 presents histograms of earnings and revenue surprises based on analyst and crowdsourced forecasts. The distribution of analyst earnings surprises is highly asymmetric, as substantially more firms just beat rather than just miss expectations (Degeorge et al. 1999; Dechow et al. 2003; Durtschi and Easton 2005). By contrast, the frequency distribution of ES_{Crowd} is more symmetric. These patterns are similar but somewhat less severe for revenue surprises. When we focus on firms that exactly meet or just beat analyst earnings expectations by one cent, we find that earnings miss crowdsourced expectations 53 percent of the time (untabulated). This result suggests that most of the firms that are typically classified as “suspect” of earnings or expectation management, because they have zero or small positive Street earnings surprises, miss the crowdsourced consensus.¹⁷

Figure 3 presents differences in the trajectories of the analyst and crowdsourced forecasts before earnings announcements. For each of the 180 days before an earnings announcement, we compute the analyst and crowdsourced consensus and plot the frequencies with which the consensus is higher (optimistic) versus lower (pessimistic) than reported earnings or revenues. Panel A reveals a strong walkdown in analyst earnings forecasts. The 95 percent confidence interval suggests that the sell-side consensus becomes significantly pessimistic between days -120 and -90 (i.e., the mean frequency is significantly lower than 0.5). This bias becomes

variable would induce a look-ahead bias. Nevertheless, our inferences remain the same when we winsorize the announcement return variable.

¹⁷For examples of studies that classify firms as “suspect” when Street earnings surprises equal 0 or 1 cent per share, see Cheng and Warfield (2005), Cohen et al. (2008), or Brochet et al. (2015).

stronger around day -90 when firms have their previous-quarter earnings announcements. For crowdsourced forecasts of earnings, we find the general walkdown and increase in pessimism around day -90 are much weaker. In Panel C, we find similar evidence of a walkdown in analyst revenue forecasts, which again weakens when based on crowdsourced forecasts.

Because some contributors on Estimize are sell-side analysts, we also repeat the analysis by splitting the crowdsourced consensus into sell-side versus other contributors. Panels B and D of Figure 3 plot the crowdsourced forecast trajectories separately for the sell-side and other contributors. We find that the patterns for sell-side contributors are very similar to those of the sell-side consensus forecast. By contrast, we find the walkdown based on non-sell-side contributors is weak and does not display the strong downward revision in forecasts around the previous-quarter earnings announcement (day -90).

Panels C and D of Table 1 provide descriptive statistics for our main dependent variable, earnings announcement returns measured over the two-day window starting on the announcement date ($BHAR_{[0,1]}$), and control variables. We control for firm size and book-to-market (e.g., Bradshaw et al. 2018) and the stock returns measured in the 60-trading day window before the earnings announcement. We split the pre-announcement stock returns into the $[-60, -6]$ and $[-5, -1]$ windows to control for measurement error in the earnings surprise variable (Brown et al. 1987; Easton and Zmijewski 1989) and short-term return reversals around earnings announcements (So and Wang 2014; Johnson and So 2018), respectively. We further control for the number of forecasts included in the sell-side and crowdsourced consensus. Panel D presents descriptives for the extended sample and shows that compared to the Estimize sample, firms in this sample are smaller, have less analyst coverage, and are associated with less pessimistic bias in forecasts.

3.3. Descriptive analysis of the incremental value of crowdsourced forecasts

Following Jame et al. (2016), who provide evidence on the incremental value of crowdsourced earnings forecasts for their Estimize sample during 2012–2013, we first provide a brief

descriptive analysis of the incremental information in crowdsourced forecasts in measuring the market expectations of earnings and revenues. We extend their analysis by analyzing a larger cross-section of firms during a longer time period (2012–2020), by examining the incremental information in revenue forecasts, and by testing whether crowdsourced forecasts have incremental value beyond a “naive” forecasting strategy.

Panel A of Table 2 presents ERC estimates for Street and crowdsourced earnings and revenue surprises. To provide a clean interpretation of the differences in coefficients, to limit the effects of outliers, and to account for the nonlinear relation between earnings surprises and announcement returns (e.g., [Freeman and Tse 1992](#)), we rank the independent variables into quarterly decile portfolios and rescale the decile ranks to values between -0.5 and $+0.5$ (e.g., [Livnat and Mendenhall 2006](#)). The significant coefficients on both ES_{Street} and ES_{Crowd} in the first column confirm the incremental information in crowdsourced forecasts. In the second column, we add revenue surprises and similarly find that crowdsourced revenue forecasts incrementally explain the market reactions to earnings announcements.

An alternative explanation for the significant coefficients on the crowdsourced surprise variables is that (i) the market already debiases analyst forecasts based on other publicly available information and that (ii) Estimote contributors use this same public information to construct a debiased version of the analyst consensus that does not incorporate any new private information. This naive forecasting strategy is a reasonable possibility since Estimote presents the analyst (“Wall Street”) consensus before contributors provide their forecasts and it displays the past consensus forecasts of analysts and the crowd relative to actual performance (see Figure 1).

We therefore include a debiased version of analyst-based earnings and revenue surprises in the third column in Panel A of Table 2. We adjust the consensus analyst earnings and revenue forecasts for the median bias in previous quarters’ forecasts, similar to a naive forecasting strategy. The results in the third column illustrate that adding surprise variables based on such debiased forecasts ($ES_{BiasAdj}$ and $SS_{BiasAdj}$) causes the coefficients on the Street

earnings and revenue variables to substantially decline. The higher coefficients for $ES_{BiasAdj}$ and $SS_{BiasAdj}$ suggest these surprise measures better capture market-assessed earnings news than unadjusted analyst-based surprises. Most importantly, the incremental explanatory power of crowdsourced forecasts remains after adding the debiased surprise measures.

A remaining issue is that, while we measure surprises relative to the actual performance reported by I/B/E/S, analysts and Estimize contributors might forecast performance on a different basis. For example, Estimize might define earnings as excluding stock-based compensation expense, while the I/B/E/S majority rule may cause “actual” earnings to include stock-based compensation expense. In the fourth column of Panel A of Table 2, we restrict the analysis to cases where the I/B/E/S and Estimize EPS actuals match to two decimal places (Jame et al. 2016) and the revenue actuals are within \$1 mln, to alleviate concerns that differences in the definition of the performance metrics explain the incremental explanatory power we document. Using this alternative sample, we find our estimates remain largely unchanged, which reinforces the incremental explanatory power of the crowdsourced surprise measures for the market reactions to earnings announcements.

Having verified the incremental information in crowdsourced forecasts, we examine the well-known announcement return premia associated with meeting analyst expectations (Bartov et al. 2002; Kasznik and McNichols 2002) and condition on whether the firm also meets crowdsourced expectations. To ensure our results are not driven by differences in the definition of the performance metrics, we focus on the firm-quarters for which the actuals are within the ranges specified above. In the first column of Panel B in Table 2, the coefficient on MBE_{Street} indicates that firms that meet both analyst and crowdsourced expectations have announcement returns that are 5.9 percent higher compared to those that miss analyst expectations. This premium drops by 3.2 percent when earnings miss crowdsourced expectations.

The second column controls for Street earnings and revenue surprises to ensure these different market reactions do not capture differences in the magnitude of the surprise for

firms that meet versus miss crowdsourced expectations. The reason is that firms with a Street earnings surprise of, for example, \$0.01 are more likely to simultaneously miss the higher crowdsourced consensus forecast than firms with a Street earnings surprise of \$0.10 or \$0.20. We find that after controlling for Street earnings and revenue surprises, the return premium associated with meeting both Street and crowdsourced expectations of 1.6 percent drops to an insignificant 0.2 percent when earnings miss crowdsourced expectations. In the third and fourth columns, we include indicators for meeting revenue expectations and similarly find that the premia associated with meeting expectations drop when firms miss the crowdsourced consensus.

In Panel C of Table 2, we repeat these analyses using the debiased measures of Street earnings and revenue surprise and find similar results. Overall, these tests suggest that crowdsourced consensus forecasts capture more information than simple corrections to analyst forecasts based on public information, and that this incremental information has material implications for the stock returns around earnings announcements.

4. Main tests and results

This section discusses our main empirical tests and results. We initially focus on Estimize firm-quarters and examine whether investors price the predictive information in crowdsourced forecasts about bias in earnings and revenue surprises. We then exploit the staggered addition of firms to the Estimize platform, which allows for stronger causal inference. Using this staggered design, we first examine whether crowdsourced coverage changes the mispricing of predictable bias in earnings and revenue surprises. Next, we examine the consequences of crowdsourced coverage for average earnings announcement returns.

4.1. Pricing of the predictive value of crowdsourced forecasts

The comparatively weaker pessimistic bias in crowdsourced forecasts implies that before earnings announcements, the difference between crowdsourced and analyst forecasts can con-

tain information about the expected bias in analyst-based earnings and revenue surprises. For example, consider an analyst consensus EPS forecast of \$0.18 per share and a crowdsourced consensus of \$0.20 per share for the same firm-quarter. If the forecast difference credibly signals an expected bias in the analyst consensus of \$-0.02, the firm is expected to report a positive Street earnings surprise. Investors are then expected to condition their reactions to the earnings announcement on the expected value of a positive bias in the Street earnings surprise of \$0.02 per share.

In Table 3, we create measures of the pre-earnings announcement difference between crowdsourced and analyst forecasts and evaluate their predictive value for subsequent earnings and revenue surprises. A more positive forecast difference indicates a higher likelihood that a firm meets analyst expectations due to a biased analyst consensus, which implies that investors should discount positive surprises relatively more. If investors fully understand and price these signals, the forecast difference should not be associated with announcement returns. If, however, investors do not fully understand and process these signals (e.g., investors expect a bias of \$0.01 instead of \$0.02 in our example), we expect a positive association between the forecast difference and subsequent earnings announcement returns.

Panel A of Table 3 presents meet/beat frequencies for several bins of the difference between the crowdsourced and analyst consensus of EPS ($FD^{EPS} = F_{Crowd}^{EPS} - F_{Street}^{EPS}$). We find a strong positive relation between the forecast difference and the likelihood that firms meet analyst expectations. For example, firms are 12.5 percentage points more likely to meet analyst expectations when $FD^{EPS} = 0.01$ than when $FD^{EPS} = -0.01$. This result suggests that the forecast difference has substantial predictive power for the sign of earnings news when the analyst consensus is used to derive this news. We draw similar inferences in Panel B when we focus on the revenue forecast difference (FD^{SAL}). In both Panels A and B, we find no clear association between the forecast differences and earnings announcement returns.

Panel C of Table 3 presents regression results. For earnings forecasts (FD^{EPS}), we

focus on forecast differences between -3 and +3 cents per share (excluding differences of 0 cents) and examine the incremental frequency of meeting analyst expectations for firm-quarters in which the crowdsourced consensus exceeds the analyst consensus. We repeat the analysis for revenue forecasts (FD^{SAL}) but focus on forecast differences between -3 and 3 percent (rounded). We control for basic firm characteristics and meet/beat frequencies in the previous 12 quarters (e.g., [Ma and Markov 2017](#)). The latter control is important because our objective is to establish the incremental predictive information in the crowdsourced forecast over the Street surprise history, which is available through traditional financial information platforms.

The regression results confirm the predictive information in the forecast differences for Street earnings and revenue surprises. We find no significant association between the forecast differences and earnings announcement returns, which is consistent with investors debiasing the Street earnings surprise based on the information in the crowdsourced consensus. The only variable that significantly predicts announcement returns is the crowdsourced coverage variable (N_{Crowd}), which suggests a higher number of Estimote contributors is associated with lower subsequent earnings announcement returns.¹⁸ Based on Table 3, we conclude that conditional on crowdsourced coverage, investors efficiently price the predictive value of the crowdsourced consensus forecast for the bias in Street earnings and revenue surprises.

4.2. *Crowdsourced forecasts and the mispricing of analyst forecast bias*

An alternative explanation for the results in the previous section could be that the firms covered by Estimote are not subject to the mispricing of predictable forecast bias for reasons other than crowdsourced coverage. In the following sections, we therefore exploit the addition of firms to the Estimote platform ([Jame et al. 2022](#)) and study the dynamics of the pricing of forecast bias by additionally examining firms before their coverage by Estimote and a group

¹⁸In untabulated tests, we find the significant negative coefficient on N_{Crowd} persists after controlling for ex-post earnings and revenue surprises. This result suggests that the higher the number of contributors, the stronger the discounting of earnings announcement news.

of firms that are never covered by Estimate.

For these tests, we employ an alternative pre-earnings announcement measure of expected bias in analyst forecasts that is available for a larger cross-section of firms. This measure is based on rolling time-series of past consensus and individual analyst forecast errors and captures differences in firms' propensity to meet sell-side expectations. [Veenman and Verwijmeren \(2018, 2022\)](#) find that this measure is positively associated with the likelihood that firms meet expectations *and* with earnings announcement returns, which suggests that investors do not completely unravel the bias in Street earnings surprises. We apply the same method to capture expected bias in analyst revenue forecasts.

In Panel A of Table 4, we tabulate the average meet/beat frequencies and announcement returns for quintile portfolios formed based on expected forecast pessimism. Both the earnings ($Pess_{EPS}$) and revenue ($Pess_{SAL}$) measures identify firms that are more versus less likely to meet analyst expectations. Consistent with mispricing ([Veenman and Verwijmeren 2018](#)), firms more likely to meet earnings expectations are associated with significantly higher announcement returns. We additionally find that expected pessimism in *revenue* forecasts is positively associated with announcement returns, which suggests that the market misprices the predictable bias in both earnings and revenue surprises.¹⁹

For firms covered by Estimate, we find evidence of mispricing only in the pre-coverage period. For example, we find a statistically and economically significant announcement return spread of 68 basis points between firms with high and low probabilities of meeting earnings expectations before Estimate coverage (t -value of 5.87). This significant return spread is consistent with investors mispricing predictable forecast bias. This mispricing, however, disappears after Estimate coverage. We find similar results for sorts based on expected bias in revenue forecasts.²⁰

¹⁹In untabulated tests, we find the positive return spreads associated with predictable bias in revenue surprises are incremental to the return spreads associated with predictable bias in earnings surprises.

²⁰An alternative explanation for these results is that the mispricing weakens because the signal in the return predictor becomes weaker. In untabulated tests, we find no evidence to support this alternative explanation. We examine the average meet/beat frequencies across the quintiles for firms before and after Estimate coverage and find that for $Pess_{EPS}$, the difference in average frequency between firms in Q1 and Q5

The results in Panel A of Table 4 also suggest that, on average, Estimote firms earn positive abnormal announcement returns in the period before coverage (31.9-32.6 basis points), while this announcement premium disappears after coverage (6.2-6.5 basis points).²¹ We conjecture that this result is explained by earnings announcement premia that are driven by bias in earnings and revenue surprises (Johnson et al. 2020), and crowdsourced coverage improving investors’ assessments of this bias. We examine this conjecture in the next section.²²

Panel B of Table 4 examines the changes in mispricing using a regression framework. As in Panel A, we identify mispricing via a positive association between ex-ante forecast pessimism and earnings announcement returns. We then interact the pessimism variables with an indicator variable for firms covered by Estimote (*Estimate*) and an indicator variable for the firm-quarters after Estimote coverage (*Post*). The coefficient on the main effect ($Pess_{EPS}$ or $Pess_{SAL}$) captures the mispricing of firms never covered by Estimote. The coefficient on the interaction with *Estimate* captures the difference in mispricing between (i) Estimote-covered firms before coverage and (ii) firms that are never covered. The coefficient on the interaction with *Post* captures the change in mispricing after Estimote coverage.

The results suggest there is significant mispricing for firms that are never covered on Estimote, while the insignificant coefficients on the interaction with *Estimate* suggest the mispricing for Estimote firms before coverage is not different from the mispricing of non-covered firms. Most importantly, the significant negative coefficients on the interaction with *Post* suggest a reduction in mispricing after Estimote coverage. We find no evidence of mispricing of earnings surprises following Estimote coverage, as indicated by the sum of the

changes from 36.3 percent to 39.2 percent. This change suggests that the signal actually becomes stronger after Estimote coverage. For $Pess_{SAL}$, we similarly find that the difference in average frequency between firms in Q1 and Q5 increases from 31.3 percent to 36.6 percent.

²¹In untabulated tests, we do not find evidence that this lower announcement premium is caused by Estimote firms performing worse in terms of asset-scaled earnings or loss frequency after coverage.

²²Note that the negative unconditional average announcement returns of 2.8-3.7 basis points for all firm-quarters are even lower than the post-Estimote averages. We find (untabulated) that this lower average announcement return can be attributed to the inclusion of firms not covered by Estimote, which are smaller and perform significantly worse than firms covered by Estimote.

main and interaction effects. The mispricing of bias in revenue surprises remains statistically significant. Based on a combined measure of earnings and revenue forecast pessimism in the final regression column, we find that the combination of information in past earnings and revenue forecast errors further increases the return spreads for non-covered firms. As before, we find a significant decline in mispricing after Estimize coverage, which is consistent with our thesis that the availability of the crowdsourced alternative benchmark facilitates investors’ debiasing of earnings and revenue surprises.

A limitation of these tests is that firms’ coverage on Estimize is not random and firms added to Estimize have materially different characteristics than other firms. For example, the firms added in 2012 are relatively large growth firms with more pessimistically biased earnings forecasts (see [Jame et al. 2022](#), Table 3). In untabulated tests, we follow [Jame et al. \(2022\)](#) and match Estimize firms, in the quarter they first appear on the platform with a crowdsourced consensus, with non-Estimize firms in the same quarter. Consistent with [Jame et al. \(2022\)](#), we find firms covered by Estimize are larger and have lower book-to-market ratios, higher sell-side coverage, higher share turnover, smaller earnings surprises, and more pessimistic consensus earnings forecasts. In addition, Estimize coverage is positively associated with return volatility ([Jame et al. 2016](#)), average announcement returns before coverage, and institutional ownership. The results from estimating the mispricing of forecast bias for the eight quarters before and after Estimize coverage for the matched sample are consistent with our previous results. We find significant mispricing of earnings and revenue forecast bias for Estimize firms in the pre-coverage period but not in the post-coverage period. For matched control firms, we find evidence of mispricing in both the pre- and post-periods.

4.3. Effects on average earnings announcement returns

The descriptive statistics in Panel A of Table 4 suggest that the mispricing of analyst forecast bias before crowdsourced coverage is associated with positive abnormal earnings announcement returns. This result relates to [Johnson et al. \(2020\)](#), who conclude that

earnings surprise bias explains why firms earn positive abnormal returns around earnings announcements, also known as the earnings announcement premium (e.g., [Cohen et al. 2007](#)). In this section, we further examine the link between investors’ assessments of forecast bias and earnings announcement premia by examining the effects of crowdsourced coverage on average earnings announcement returns.

Crowdsourced coverage can affect average earnings announcement returns through two channels. First, if the crowdsourced forecast benchmark helps investors form better priors about analyst forecast bias, average announcement returns should decline because of increased discounting of positive earnings and revenue surprises. Second, [Jame et al. \(2022\)](#) conclude that crowdsourced coverage is associated with reductions in analyst forecast bias. These reductions in forecast bias should lead to lower earnings and revenue surprises and hence, holding constant investors’ ability to anticipate forecast bias, lower announcement returns.

Separating the two channels is empirically challenging. However, we can control for reductions in forecast bias by examining whether crowdsourced coverage is associated with reduced earnings announcement returns after conditioning on the levels of earnings and revenue surprises. Given this discussion, we conduct two sets of tests. First, we extend [Jame et al. \(2022\)](#) by examining the effects of crowdsourced coverage on different measures of earnings and revenue surprise. Second, we examine the effects of crowdsourced coverage on average earnings announcement returns, both before and after conditioning on analyst-based earnings and revenue surprises.

We use a generalized difference-in-differences design that exploits the staggered addition of firms to Estimote.²³ Different from Section 4.2, in which we examine the change in the *association* between announcement returns and ex-ante measures of forecast pessimism, we now focus on changes in the *level* of our outcome variables. This design allows us to better isolate the effect of Estimote coverage using a two-way fixed-effects structure with firm and

²³We address the parallel trends assumption and potential concerns about staggered difference-in-differences in two-way fixed effects estimations in Section 5.3

year-quarter fixed effects. Given the staggered addition of firms to Estimize during the 20 quarters in 2012–2016, our tests quantify the difference-in-differences in announcement returns after crowdsourced coverage relative to firms that are not (yet) covered on Estimize.

The results in Table 5 suggest that crowdsourced coverage is associated with significant reductions in the frequency with which firms just meet analyst earnings expectations, meet or just meet revenue expectations, and meet both earnings and revenue expectations.²⁴ For example, relative to firms not (yet) covered, crowdsourced coverage is associated with a 3.2 percentage point decline in the frequency of meeting both earnings and revenue expectations. This is a material decline given the unconditional frequency of 44.8 percent for meeting both forecasts (untabulated). These results extend those of [Jame et al. \(2022\)](#) for alternative measures of earnings and revenue surprise and confirm that reductions in forecast bias might cause reductions in earnings announcement premia.²⁵

Using the same research design in Table 6, we find that crowdsourced coverage significantly reduces earnings announcement returns. On average, announcement returns drop by about 50 basis points in the post-coverage period. We find that this result is robust to controlling for the levels of earnings and revenue surprises, as well as to conditioning the analysis on the sign of the surprise. Thus, these results suggest that even after accounting for the effects of the reduced-forecast-bias channel on announcement returns, the lower average announcement returns reflect investors’ improved information processing of forecast bias. For revenue surprises, we find the reduction in average announcement returns is concentrated only in the subset of firms that meet revenue expectations, which are the observations where bias in analyst revenue forecasts is most likely. Overall, we conclude that these results are consistent with the notion that increased visibility of forecast bias,

²⁴To address potential concerns about bias induced by the inclusion of time-varying control variables in fixed-effect difference-in-differences estimations ([Sant’Anna and Zhao 2020](#)), we separately present results of tests that exclude and include control variables in Panels A and B, respectively.

²⁵Although the weakly significant effect on MBE_{Street} in Panel A and insignificant effect in Panel B appear inconsistent with [Jame et al. \(2022\)](#), who report in their robustness tests (Table 6) a significant reduction in meet/beat frequency for firms added to Estimize in 2012, we find a more substantial and significant reduction when we focus solely on firms added to Estimize in 2012 and/or 2013.

as a result of crowdsourced coverage, improves investor information processing and reduces earnings announcement premia.

5. Additional analyses

5.1. Managerial responses to crowdsourced coverage

The existence of an additional, less pessimistic, performance benchmark may place additional pressure on managers to take actions to meet both analyst and crowdsourced expectations (Ott et al. 2020; Sul 2020). At the same time, the reductions in earnings announcement premia (our study) and forecast pessimism (Jame et al. 2022) following crowdsourced coverage can reduce the net benefits to managers of using costly earnings management or guidance strategies to meet expectations. In Figure 4, we illustrate how the complex dynamics of the potential effects of forecast bias exposure on managers' strategic behavior. The causal diagram suggests it is unclear how crowdsourced coverage affects earnings management and guidance. We empirically examine these outcomes in this section.

5.1.1. Accrual-based earnings management

If managers respond to the pressure of the additional performance benchmark by using income-increasing accruals management to meet expectations, firms that meet both forecast benchmarks should be associated with higher abnormal accruals than firms that meet Street expectations only. We examine this prediction using the following model:

$$WCA_{it} = \beta_X MBE_{X,it} + \sum_{j=1}^J \mathbb{1}_j \left(\gamma_{0j} + \left[\sum_{k=2}^4 \gamma_{1j}^k QTR_{it}^k \right] + \gamma_{2j} \Delta S_{it} + \gamma_{3j} WCA_{it-4} \right. \\ \left. + \left[\sum_{k=2}^5 \gamma_{4j}^k ROA_Dum_{it}^k \right] + \left[\sum_{k=2}^5 \gamma_{5j}^k SG_Dum_{it}^k \right] + \left[\sum_{k=2}^5 \gamma_{6j}^k MTB_Dum_{it-1}^k \right] \right) + \varepsilon_{it}. \quad (1)$$

Following Collins et al. (2017), we model quarterly working capital accruals (WCA) based on fiscal-quarter indicators (QTR), the adjacent-quarter change in revenues (ΔS), lagged working capital accruals, and nonlinear functions of performance (ROA), seasonally-adjusted sales growth (SG), and growth (MTB). ΔS is measured either as the basic change

in revenues (“Jones”) or the change in revenues adjusted for the change in receivables (“Mod-Jones”). Following [Chen et al. \(2018\)](#), we define abnormal accruals as the level of accruals that is not explained by the determinants interacted with two-digit SIC industry-year indicators (denoted $\mathbb{1}_j$) in a one-step estimation. After controlling for these interacted determinants, the coefficient on the variable of interest (MBE_X) captures the incremental abnormal accruals for firms that meet the performance benchmark.

Panel A of Table 7 presents the results. When included individually, the positive and significant coefficients on the indicator variables for meeting Street (MBE_{Street}) and crowdsourced expectations (MBE_{Crowd}) suggest that firms that meet expectations have higher levels of abnormal working capital accruals. When we replace MBE_{Crowd} with an indicator for whether firms meet both crowdsourced and Street expectations, the positive and significant coefficients on MBE_{Both} suggest that compared to meeting Street expectations only, firms that meet both performance benchmarks are associated with significantly higher abnormal working capital accruals.

In Panel B of Table 7, we examine how managers respond in quarters after firms miss the crowdsourced consensus. We replace MBE_X in equation (1) with two indicator variables that capture whether the firm met Street expectations in quarter $t - 1$ and whether it simultaneously missed crowdsourced expectations, respectively. We find that, relative to meeting both Street and crowdsourced expectations, firms have significantly higher levels of abnormal accruals in quarters after they miss crowdsourced expectations. This result suggests managers respond to the adverse consequences of missing crowdsourced expectations by inflating accruals in the following quarter.

In Panel C of Table 7, we expand the sample to non-Estimize firm-quarters and interact MBE_{Street} with *Estimize* and *Post*, similar to Table 4. We examine whether firms that meet analyst expectations use additional income-increasing accruals in response to (i) the higher (i.e., less pessimistic) additional forecast benchmark and (ii) the higher analyst consensus after crowdsourced coverage. Although we find a positive association between abnormal

accruals and the meeting of Street earnings expectations for this sample, the insignificant coefficient on $MBE_{Street} \times Post$ suggests that firms do not change their use of accrual management to meet Street expectations after crowdsourced coverage. The significant negative coefficient on $MBE_{Street} \times Estimate$ suggests that firms covered by Estimate make less use of accrual management to meet expectations both before and after coverage.

Overall, these results suggest that while firms that meet both analyst and crowdsourced expectations have elevated levels of abnormal accruals, the coverage of firms on Estimate is not associated with a change in the use of accrual management to meet expectations.

5.1.2. *Earnings and revenue guidance*

In Table 8, we examine whether Estimate coverage is associated with changes in guidance. Using a design similar to Tables 5 and 6, we find that, relative to firms never covered by Estimate, Estimate firms experience significant increases in their propensity to provide earnings or revenue guidance. Most importantly, we find that these increases in guidance frequency are driven by downward guidance. For example, in Panel A we find that Estimate firms are associated with an increase in the frequency of providing downward revenue guidance of 3.8 percentage points. The effects are material and found both without and with control variables included in Panels A and B, respectively. These results suggest that managers use public guidance in attempts to steer expectations downward after the crowdsourced consensus forecast becomes available for their firms, which is less pessimistically biased than the traditional analyst consensus forecast.

5.2. *The role of investor attention to firms on the Estimate platform*

Our earlier discussion and Appendix A suggest that the market primarily focuses on the sell-side analyst consensus as the performance benchmark. Given this fixation, a necessary condition for crowdsourced coverage to affect investor information processing is that at least some of investors' attention is redirected towards the crowdsourced consensus. We therefore

exploit unique data on [estimize.com](https://www.estimize.com) page visits, which allows us to examine the role of firm-specific investor attention on the Estimize platform on a particular day.²⁶

We construct a measure of abnormal investor attention to crowdsourced forecast information around earnings announcements based on ticker-specific web traffic on [estimize.com](https://www.estimize.com). We measure the ratio of the average daily viewing activity from day $t - 2$ through day $t + 2$ to the average daily viewing activity from day $t - 11$ to day $t - 50$, where day t is the earnings announcement date. We focus on unique daily user visits by eliminating duplicate observations for any user-firm-day combination and ignore web traffic that is caused by contributors providing forecasts. For each of the firms that receive Estimize coverage, we then measure the average of abnormal attention across all earnings announcements after coverage and split covered firms into those that receive high versus low abnormal attention.

In untabulated tests, we re-estimate our two-way fixed effects difference-in-differences regressions by setting *Post* equal to 1 only for firms that receive high or low abnormal attention on Estimize around earnings announcements. We find that our results of lower earnings announcement premia, increased downward earnings and revenue guidance, and lower frequency of meeting analyst earnings and revenue forecasts, are significant only for those firms that receive high average abnormal attention around earnings announcements. These results lend further support to our conclusion that crowdsourced coverage affects investors' information processing around earnings announcements.²⁷

5.3. *Additional sensitivity tests*

The identifying assumption underlying our difference-in-differences analyses is that the trends in the outcome variables would have moved in tandem in the counterfactual situation

²⁶We thank Estimize for kindly sharing the data with us, which they collected between January 1, 2013 and July 1, 2019.

²⁷These tests and results should be interpreted with the caveat in mind that the viewing activity of a firm around a particular earnings announcement on [estimize.com](https://www.estimize.com) is unlikely to be random. Although we average the abnormal attention across announcements for each firm, this is unlikely to fully resolve potential endogeneity concerns. Also, note our variable measures investor attention conservatively since it does not capture views of Estimize forecasts through other sources, such as Bloomberg terminals.

that Estimote had not added a firm to the platform. Although we cannot formally test this assumption, an alternative solution is to explore whether the treatment (Estimote) and control (non-Estimote) firms exhibit similar pre-trends. Figure 5 presents results from a dynamic fixed effects estimation in which we examine how the treatment effect varies in event-time. Following Baker et al. (2022), we treat the pre-coverage year as the baseline and include separate indicators that pick up the difference-in-differences estimates in years -3, -2, 0, +1, +2, and +3. In support of the parallel trends assumption, we find no evidence that suggests the pre-trends are dissimilar for meet/beat incidence, announcement returns, and downward guidance frequency. Instead, we find significant changes for each of the four variables as of year +1. We conclude that our results are unlikely driven by differences in pre-trends that continue after the start of Estimote coverage.

Recent advances in the econometrics literature further suggest that inferences based on two-way fixed effects designs with staggered treatment timing can be biased when treatment effects are dynamic and heterogeneous (e.g., de Chaisemartin and D’Haultfoeuille 2020; Callaway and Sant’Anna 2021; Goodman-Bacon 2021; Sun and Abraham 2021). To address this issue, we follow Cengiz et al. (2019), Dey et al. (2021), and Baker et al. (2022) by estimating stacked event-by-event regressions.²⁸ Table 9 shows that our results on the reductions in announcement returns and meet/beat frequency, and increases in downward earnings and revenue guidance, are robust to using this alternative design. We therefore conclude that concerns over staggered difference-in-differences designs are unlikely to affect our inferences.

²⁸In this design, we treat each of the 20 calendar quarters during 2012–2016 in which firms are added to Estimote as a cohort and create 20 separate datasets. The treatment firms in a cohort are those added to Estimote in that quarter and control firms are those not included on Estimote during the event window from 8 quarters before to 8 quarters after the cohort-quarter. We next create a stacked dataset by pooling the 20 event-specific “clean 2×2 ” datasets and aligning them in event-time. For each of the datasets, we create separate fixed effects such that the firm and year-quarter fixed effects are now cohort-firm and cohort-year-quarter fixed effects (Baker et al. 2022). In untabulated tests, we find our inferences are similar when we define the control group more restrictively using firms that are never added to Estimote during the sample period 2012–2020.

6. Summary and conclusion

Although analysts' short-term forecasts tend to be pessimistically biased, these forecasts are typically used to determine the news in an earnings announcement. Using crowdsourced forecasts from Estimote, we examine whether these alternative forecasts of earnings and revenues help investors unravel the predictable bias in earnings announcement news.

We first show that the return premium for meeting or beating analyst expectations is eliminated when firms miss the crowdsourced forecast benchmark. Next, using the pre-earnings announcement difference between crowdsourced and analyst consensus forecasts, we find that investors understand and price the predictive information in crowdsourced forecasts for the bias in analyst-based earnings and revenue surprises. Exploiting the staggered addition of firms to the Estimote platform over time, we find significant reductions in the mispricing of bias in analyst earnings and revenue forecasts and significant reductions in earnings announcement premia. We find some evidence of income-increasing accruals for firms that meet crowdsourced expectations and show that managers respond to crowdsourced coverage through downward earnings and revenue guidance.

We conclude that user-generated content on crowdsourced financial information platforms can help investors discount biases in traditional sources of equity research and better process information about the news in an earnings announcement. We emphasize, however, that although we find improvements in investors' information processing and reductions in analyst forecast bias, the pessimistic bias in earnings and revenue forecasts remains an important concern. In each of our sample years, we find that Estimote-covered firms still meet or beat analyst earnings expectations in over 70 percent of firm-quarters. Finally, a caveat of our study is that the coverage of firms on the Estimote platform is not random and plausibly related to concerns over the capital market effects of biases in sell-side analyst research. Therefore, our results may not generalize to randomly selected firms for which biases in analyst research are less prominent.

Appendix A. Illustration of consensus EPS forecast and surprise information displayed on prominent financial information platforms

The table below presents a summary of the consensus forecasts and earnings surprise information that is displayed on prominent financial information platforms and media for Apple Inc. (AAPL) on July 6, 2022. Quarter 2022Q3 (exp) refers to the to-be-reported fiscal quarter ending June 30, 2022. Quarters 2020Q3 through 2022Q2 refer to the most recently reported eight fiscal quarters. Each line that starts with the quarter-label contains the consensus EPS forecast displayed on a particular platform. The shaded green lines display the measures of percentage earnings surprise that are displayed by the platforms.

For comparison purposes, the first column displays the Estimate consensus. The second column displays the consensus forecast available from the I/B/E/S summary history files. For Apple’s 2022Q3, this is the monthly consensus (mean) forecast computed by I/B/E/S on June 16, 2022. For the remaining quarters this is the latest monthly consensus before the earnings announcement. The remaining columns display the information provided through Bloomberg terminals, Factset terminals, NASDAQ.com, CNBC, and Yahoo! Finance. The platforms vary in the length of the history of earnings surprise information they provide. Earnings beats vs. misses are always highlighted by green vs. red text. Untabulated analyses suggest consensus forecast and earnings surprise information is highly similar when presented through other platforms and media, such as WSJ.com, SeekingAlpha, Google, MarketBeat, TipRanks, Investing.com, and StreetInsider. The key insight from this analysis is that consensus forecast and earnings surprise information on prominent financial information platforms and media overlaps substantially with the sell-side analyst forecasts as reflected in the I/B/E/S consensus forecasts. The Estimate consensus forecast differs from the analyst consensus forecast in each of the quarters and is less pessimistic.

When focusing on the most recently announced quarter for Apple at the time of this case study (2022Q2, announced on April 28, 2022), we find that media coverage of this earnings announcement consistently refers to the analyst consensus of 1.42 (Barrons and WSJ) or 1.43 (CNBC) per share in identifying the earnings surprise. At the same time, all platforms/media identify the reported number as being 1.52 per share. This example further illustrates the media fixation on the analyst consensus forecast in determining the news in an earnings announcement.

	Estimate	I/B/E/S	Bloomberg	Factset	NASDAQ	CNBC	Yahoo!
2022Q3 (exp)	1.30	1.16	1.152	1.16	1.14	1.16	1.16
2022Q2	1.51	1.43	1.423	1.42	1.43	1.43	1.43
			6.82%	6.7%	6.29%	6.4%	6.3%
2022Q1	1.96	1.88	1.900	1.90	1.89	1.89	1.89
			10.53%	10.6%	11.11%	11.2%	11.1%
2021Q4	1.34	1.23	1.236	1.24	1.24	1.24	1.24
			0.32%	0.1%	0%	–	0%
2021Q3	1.16	1.00	1.013	1.01	1.00	1.01	1.01
			28.33%	28.6%	30%	29.1%	28.7%
2021Q2	1.07	0.98	0.991	0.99		0.99	
			41.27%	41.9%		41.8%	
2021Q1	1.47	1.40	1.417	1.42		1.41	
			18.56%	18.6%		19.0%	
2020Q4	0.76	0.70	0.702	0.71		0.70	
			3.99%	2.8%		–	
2020Q3	0.53	0.50	0.518	0.51		0.51	
			24.52%	25.6%		26.2%	

Appendix B. Variable definitions

Variable	Description
$BHAR_{[-60,-6]}$	Size-adjusted buy-and-hold return from CRSP measured from trading day $t - 60$ through day $t - 6$ relative to the earnings announcement date. Size adjustments are based on NYSE/AMEX/NASDAQ market capitalization cutoffs.
$BHAR_{[-5,-1]}$	Size-adjusted buy-and-hold return from CRSP measured from trading day $t - 5$ through day $t - 1$ relative to the earnings announcement date. Size adjustments are based on NYSE/AMEX/NASDAQ market capitalization cutoffs.
$BHAR_{[0,1]}$	Earnings announcement returns from CRSP, measured as the size-adjusted buy-and-hold returns from CRSP on days 0 through +1 relative to the earnings announcement date. Size adjustments are based on NYSE/AMEX/NASDAQ market capitalization cutoffs.
BTM	Book-to-market ratio, measured as the ratio of book value of common equity (Compustat: CEQQ) to market value of equity (PRCCQ $\times$ CSHOQ) at the end of the fiscal quarter.
$ES_{BiasAdj}$	Bias-adjusted Street earnings surprise, where the consensus analyst forecast (F_{Street}^{EPS}) is adjusted for the median bias in analyst EPS forecasts measured over the previous 12 quarterly earnings announcement and requiring a minimum of 4 previous quarters.
ES_{Crowd}	Crowdsourced earnings surprise, measured as the difference between I/B/E/S actual earnings per share and the crowdsourced consensus forecast from Estimote in cents per share, scaled by the closing stock price from CRSP two days before the earnings announcement. The consensus (mean) forecast (F_{Crowd}^{EPS}) is constructed from Estimote contributors' most recent quarterly earnings forecasts made within 120 of the announcement date. When a contributor issues multiple forecasts for a firm on the same day and time, we use the average of these forecasts. Estimates made on the announcement date or flagged by Estimote as unreliable are excluded. When constructing the consensus, we use the split-adjustment factors provided by Estimote to adjust for stock splits between the forecast and announcement date.
ES_{Street}	Street earnings surprise, measured as the difference between I/B/E/S actual earnings per share and the sell-side consensus forecast in cents per share, scaled by the closing stock price from CRSP two days before the earnings announcement. The consensus (mean) forecast (F_{Street}^{EPS}) is constructed from analysts' most recent quarterly earnings forecasts made within 120 days of the announcement date. Analyst forecast and actual earnings data are from the I/B/E/S Unadjusted Detail History and Actual files. To account for stock splits between the forecast and earnings announcement date, we obtain cumulative stock split factors from CRSP. When the split factor on the forecast date differs from that on the earnings announcement date, the forecast of earnings per share is multiplied by the ratio of split factor on the announcement date to the split factor on the forecast date.
$Estimize$	Indicator variable set equal to 1 for firms covered by Estimote at any point in time in our sample, and 0 otherwise.

Variable	Description
FD^{EPS}	Variable capturing the difference between the crowdsourced and analyst consensus EPS forecasts before the earnings announcement ($F_{Crowd}^{EPS} - F_{Street}^{EPS}$); measured in cents per share.
FD^{SAL}	Variable capturing the relative difference between the crowdsourced and analyst consensus revenue forecasts before the earnings announcement ($F_{Crowd}^{SAL}/F_{Street}^{SAL} - 1$); measured in rounded percentages.
$Guide_Y^X$	Indicator variable based on I/B/E/S Guidance data, set equal to 1 when the firm provides guidance of earnings ($X = EPS$) or revenues ($X = SAL$), and 0 otherwise; measured separately based on whether the guidance is higher ($Y = Up$) or lower ($Y = Down$) than the prevailing consensus forecast of EPS or revenue.
MBE_{12q}	Fraction of the previous twelve quarters in which the firm meets or beats the sell-side consensus earnings forecast ($MBE_{Street} = 1$).
$MBE_{BiasAdj}$	Indicator variable set equal to 1 when earnings per share exactly meet or beat the bias-adjusted analyst consensus forecast ($ES_{BiasAdj} \geq 0$), and 0 otherwise.
MBE_{Crowd}	Indicator variable set equal to 1 when earnings per share exactly meet or beat the crowdsourced consensus forecast ($ES_{Crowd} \geq 0$), and 0 otherwise.
MBE_{Street}	Indicator variable set equal to 1 when earnings per share exactly meet or beat the sell-side analyst consensus forecast ($ES_{Street} \geq 0$), and 0 otherwise.
$MBES_{Street}$	Indicator variable set equal to 1 when both earnings per share and revenues meet or beat the sell-side analyst consensus forecast, and 0 otherwise.
MBS_{12q}	Fraction of the previous twelve quarters in which the firm meets or beats the sell-side consensus revenue forecast ($MBS_{Street} = 1$).
$MBS_{BiasAdj}$	Indicator variable set equal to 1 when revenues meet or beat the bias-adjusted analyst consensus forecast ($SS_{BiasAdj} \geq 0$), and 0 otherwise.
MBS_{Crowd}	Indicator variable set equal to 1 when revenues meet or beat the crowdsourced consensus forecast ($SS_{Crowd} \geq 0$), and 0 otherwise.
MBS_{Street}	Indicator variable set equal to 1 when revenues meet or beat the sell-side analyst consensus forecast ($SS_{Street} \geq 0$), and 0 otherwise.
$Miss_{Crowd}$	Indicator variable set equal to 1 when performance is lower than the crowdsourced consensus forecast, and 0 otherwise; when interacted with MBE_{Street} , this variable is based on missing crowdsourced earnings expectations ($ES_{Crowd} < 0$); when interacted with MBS_{Street} , this variable is based on missing crowdsourced revenue expectations ($SS_{Crowd} < 0$).
$MJBE_{Street}$	Indicator variable set equal to 1 when EPS exactly meets or just beats analyst expectations by one cent per share, 0 for negative earnings surprises.
$MJBS_{Street}$	Indicator variable set equal to 1 when revenues exactly meets or just beats analyst expectations by no more than 1 percent, 0 for negative revenue surprises.

Variable	Description
N_{Crowd}	Number of Estimize contributors with an outstanding forecast of one-quarter-ahead earnings per share up until day $t - 1$ relative to the quarterly earnings announcement date.
N_{Street}	Number of I/B/E/S sell-side analysts with an outstanding forecast of one-quarter-ahead earnings per share up until day $t - 1$ relative to the quarterly earnings announcement date.
$Pess_{EPS}$	Ex-ante measure of expected pessimistic bias in the consensus analyst EPS forecast based on Veenman and Verwijmeren (2018) : this variable is the sum of the relative frequency with which the firm beat (versus missed) consensus analyst expectations in the previous 12 quarters and the relative frequency with which any firm beat (versus missed) an individual analyst’s forecasts in the previous 12 months, averaged across the analysts forecasting earnings for the current firm-quarter.
$Pess_{SAL}$	Ex-ante measure of expected pessimistic bias in the consensus analyst revenue forecast, measured similarly to $Pess_{EPS}$ but based on past errors in revenue forecasts.
$Pess_{EPS+SAL}$	The sum of $Pess_{EPS}$ and $Pess_{SAL}$.
$Post$	Indicator variable set equal to 1 for all firm-quarters since the first quarter the firm was first included on the Estimize platform, and 0 otherwise.
$Size$	Firm size, measured as the total market value of equity at the end of the fiscal quarter (Compustat: $PRCCQ \times CSHOQ$).
$SS_{BiasAdj}$	Bias-adjusted Street revenue surprise, where the consensus analyst forecast (F_{Street}^{SAL}) is adjusted for the median percentage bias in analyst revenue forecasts measured over the (maximum) 12 previous quarterly earnings announcement.
SS_{Crowd}	Crowdsourced revenue surprise, measured similar to ES_{Crowd} but based on consensus crowdsourced forecast of revenues (F_{Crowd}^{SAL}) and scaled by market value two days before the earnings announcement.
SS_{Street}	Street revenue surprise, measured similar to ES_{Street} but based on the consensus analyst forecast of revenues (F_{Street}^{SAL}) and scaled by market value two days before the earnings announcement.
WCA	Working capital accruals, measured based on Compustat cash flow statement data ($RECCHQ + INVCHQ + APALCHQ + TXACHQ + AOLOCHQ$), and scaled by lagged total assets; this variable is used as dependent variable in equation (1), where ΔS is measured either as the change in revenues ($\Delta SALE$; “Jones”) or the change in revenues adjusted for the change in receivables ($\Delta SALE + RECCHQ$; “Mod-Jones”) scaled by lagged assets (AT), ROA_Dum are indicator variables based on quarterly quintile portfolios formed based on net income (IB) scaled by lagged assets, SG_dum are indicator variables formed based on the quarterly quintiles of seasonally-differenced percentage growth in revenues, and MTB_Dum are indicator variables formed based on the quarterly quintiles of the market-to-book ratio measured as lagged market value ($PRCCQ \times CSHOQ$) scaled by lagged book value of equity (CEQ).

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Figure 1

Illustration of information displayed on Estimote platform

Screenshot of the Estimote page of Netflix, Inc. (NFLX). Estimote presents the current sell-side (“Wall St”) consensus earnings forecasts for the upcoming four quarters. Estimote additionally provides a summary table and chart of the quarterly earnings forecasts (crowdsourced and sell-side) and actual earnings for the previous quarters. In 2017Q4, 2018Q1, 2018Q4, and 2019Q2, NFLX simultaneously meets the sell-side consensus and misses the Estimote consensus. The Estimote consensus for NFLX is higher than the sell-side consensus in each of the quarters, while the firm misses expectations in one (five) out of the eight quarters based on the sell-side (crowdsourced) consensus.

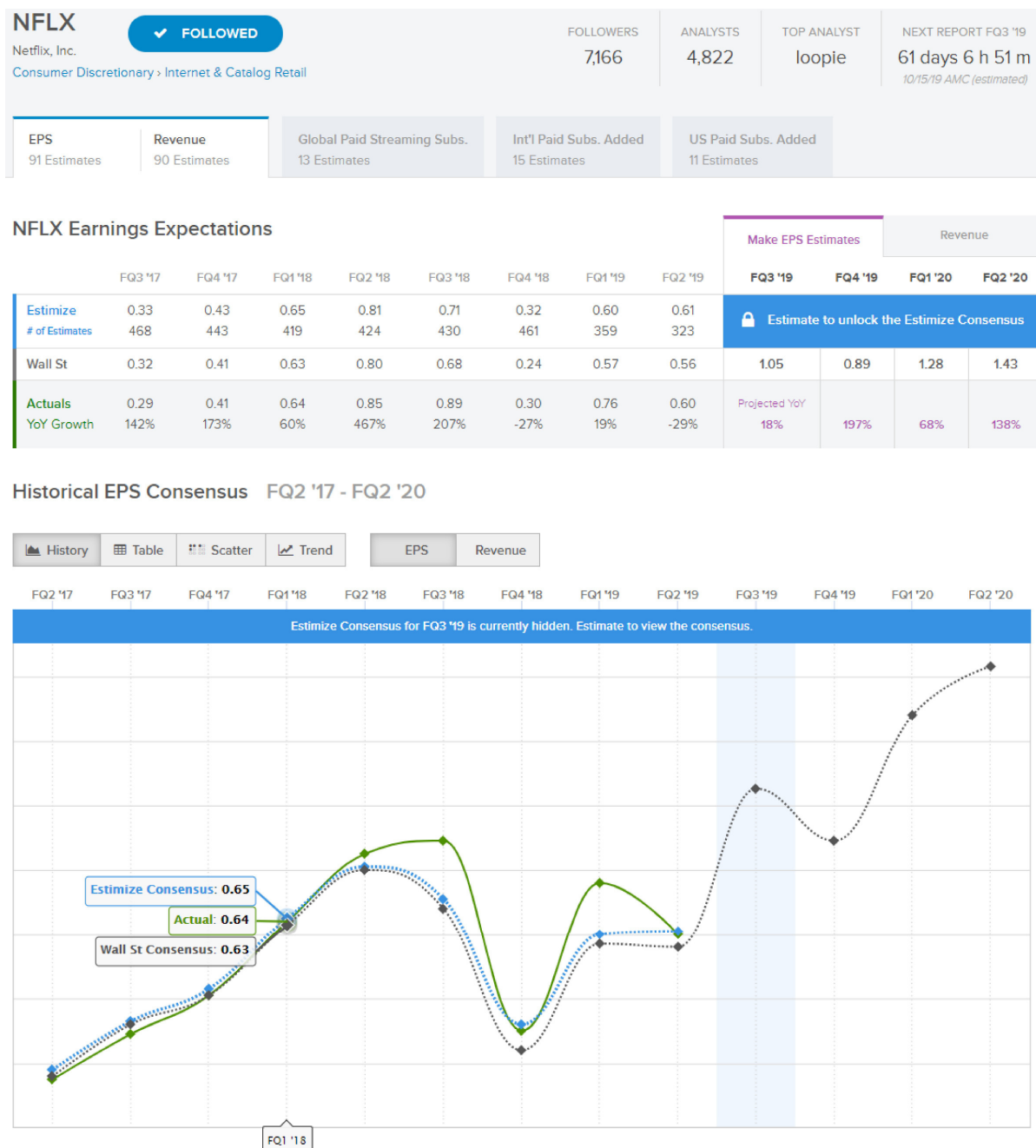
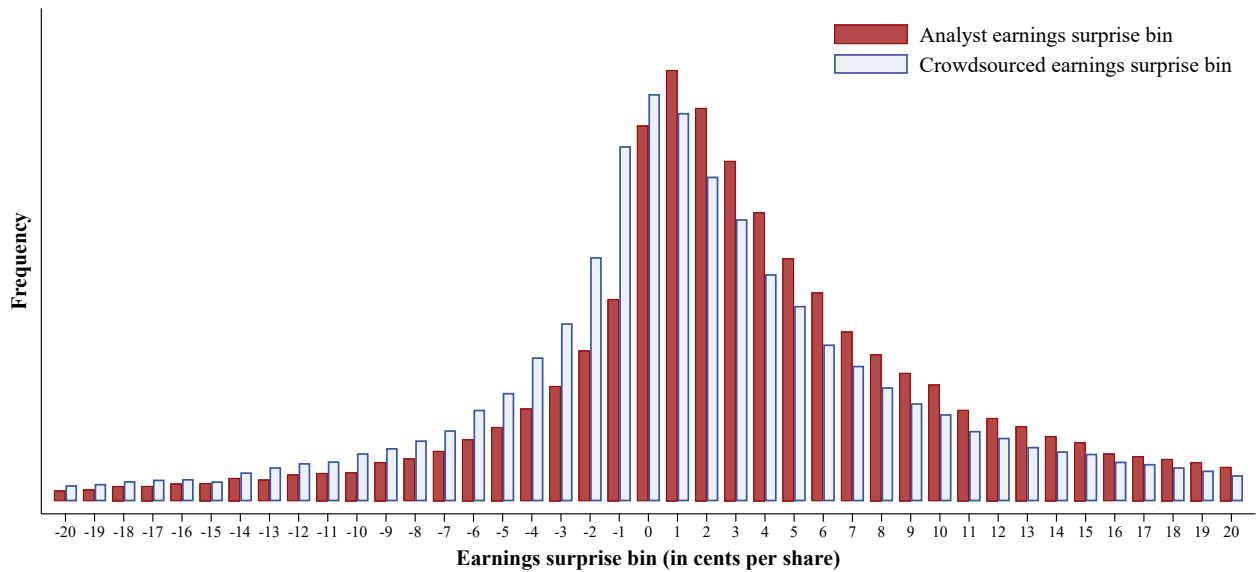


Figure 2

Earnings and revenue surprise distributions based on analyst vs. crowdsourced forecasts

This figure presents frequency distributions of earnings (Panel A) and revenue (Panel B) surprise variables constructed from analyst and crowdsourced consensus forecasts. For presentation purposes, the graphs exclude earnings (revenue) surprises exceeding the absolute value of 20 cents per share (20 percent of the forecast). The red (filled) bars capture the frequency of analyst-based surprises derived from forecasts on I/B/E/S. The light blue (shaded) bars capture the frequency of crowdsourced surprises derived from forecasts on Estimize. The horizontal axes denote the surprise bins in cents per share for earnings (Panel A) and percent of the forecast for revenues (Panel B). To ensure performance is measured on the same basis, we require actual earnings per share in I/B/E/S and Estimize to match to two decimal place ([Jame et al. 2016](#)) and require actual revenues in I/B/E/S and Estimize to be within 0.1 percent.

Panel A: Frequency distributions of earnings surprises



Panel B: Frequency distributions of revenue surprises

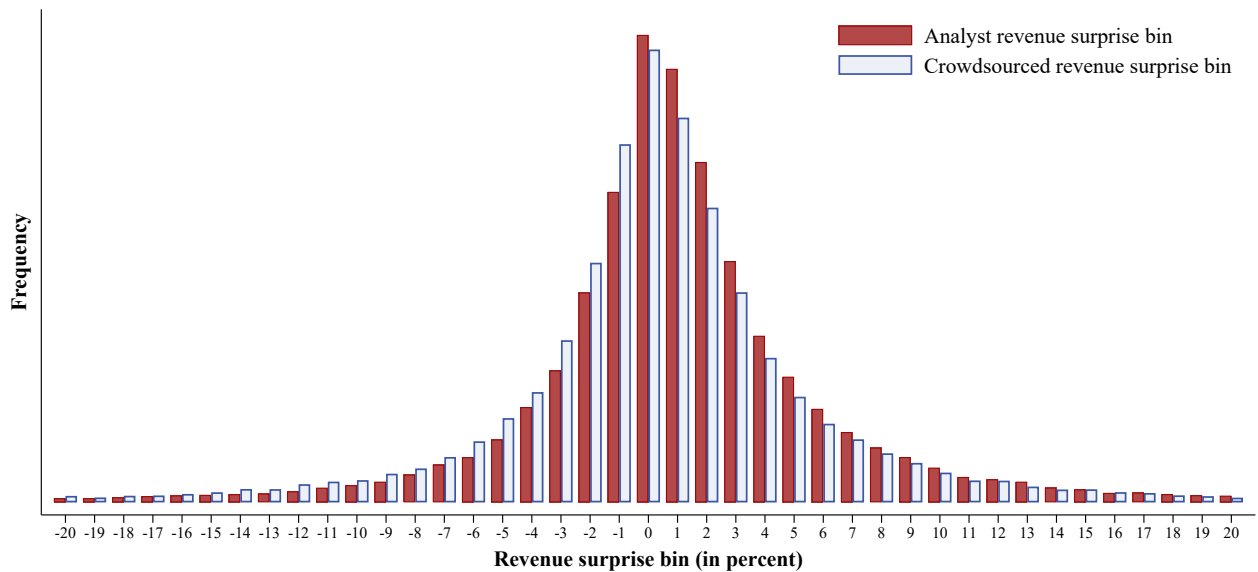


Figure 3

Forecast walkdowns based on analyst vs. crowdsourced consensus forecasts

This figure plots the forecast walkdown based on crowdsourced versus analyst consensus forecasts. Panels A and B focus on EPS forecasts, while Panels C and D focus on revenue forecasts. On each of the 180 calendar days before the earnings announcement date, we compute the analyst and crowdsourced consensus (mean) forecast using forecasts released in the preceding 120 days. On the left, the solid red (dashed blue) line depicts the daily mean analyst (crowdsourced) consensus forecast. On the right, the solid red (dashed blue) line depicts the daily mean consensus forecast for sell-side analyst contributors on Estimize (contributors on Estimize excluding sell-side analysts). The shaded area captures the 95 percent confidence interval around the mean forecast based on standard errors adjusted for clustering by firm and year-quarter. The horizontal axis depicts the calendar days relative to the earnings announcement and the vertical axis denotes the fraction of daily consensus estimates that are optimistic versus pessimistic. Values above (below) 0.5 indicate the consensus forecast is optimistic (pessimistic) relative to the actual number.

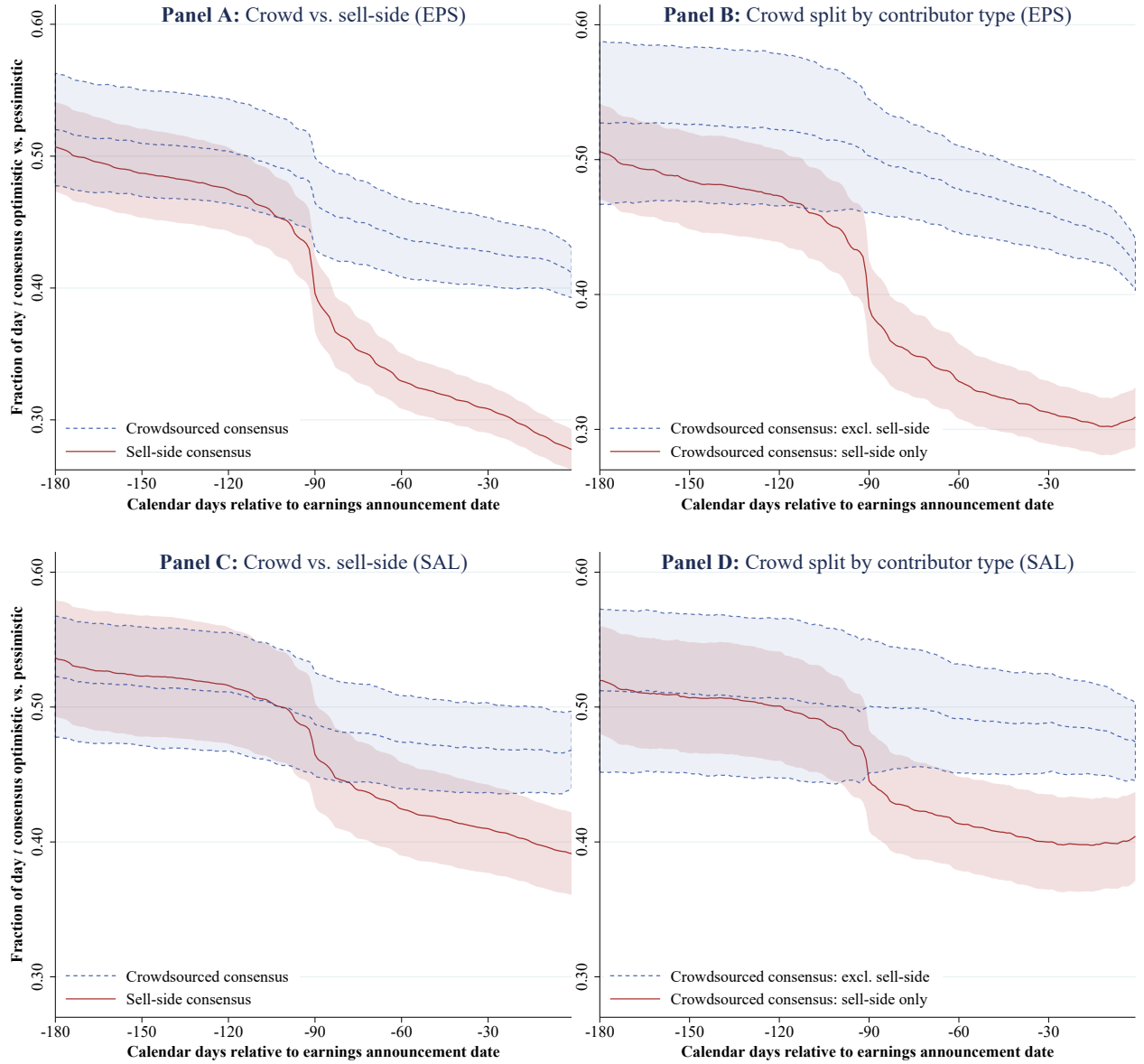


Figure 4

Conceptual predictions on managerial responses to forecast bias exposure

This figure presents directed acyclic graphs (Pearl 1995; Morgan and Winship 2015; Gow et al. 2016) to depict the conceptual linkages between the constructs of interest and to provide a visualization of the potential effects of crowdsourced coverage and forecast-bias exposure on managers. The arrowed lines capture causal chains, the solid dots indicate observable constructs, and the hollow dots indicate unobservable constructs. *Bias exposure* refers to the presence of a less biased alternative expectation benchmark than the sell-side analyst consensus. *Return premium* refers to the premium for meeting/beating analyst earnings or revenue expectations. We predict that the presence of a less biased alternative forecast benchmark increases the visibility of analyst forecast bias and reduces the return premium for meeting or beating analyst expectations. *Analyst pessimism* refers to the observed pessimistic bias in the consensus analyst forecast, which Jame et al. (2022) conclude additionally declines following crowdsourced coverage. This reduced pessimistic bias causes firms to meet/beat analyst expectations less frequently (*MBE/MBS*). *EM* refers to income-increasing earnings management, while *Guide* refers to the use of downward guidance. Given managers' expected costs of using earnings management and downward guidance, we argue that the net effects of the reduced return premiums and analyst forecast pessimism on earnings management and guidance are ex-ante unclear. The effects on earnings management depend on the expected costs of managing earnings compared to the costs associated with missing expectations based on unmanaged earnings. The effects on guidance depend on the expected negative stock price effects of providing downward guidance compared to the costs of missing unmanaged expectations. Finally, the effects of guidance on analyst forecasts depend on the extent to which analysts are willing to follow these cues from managers.

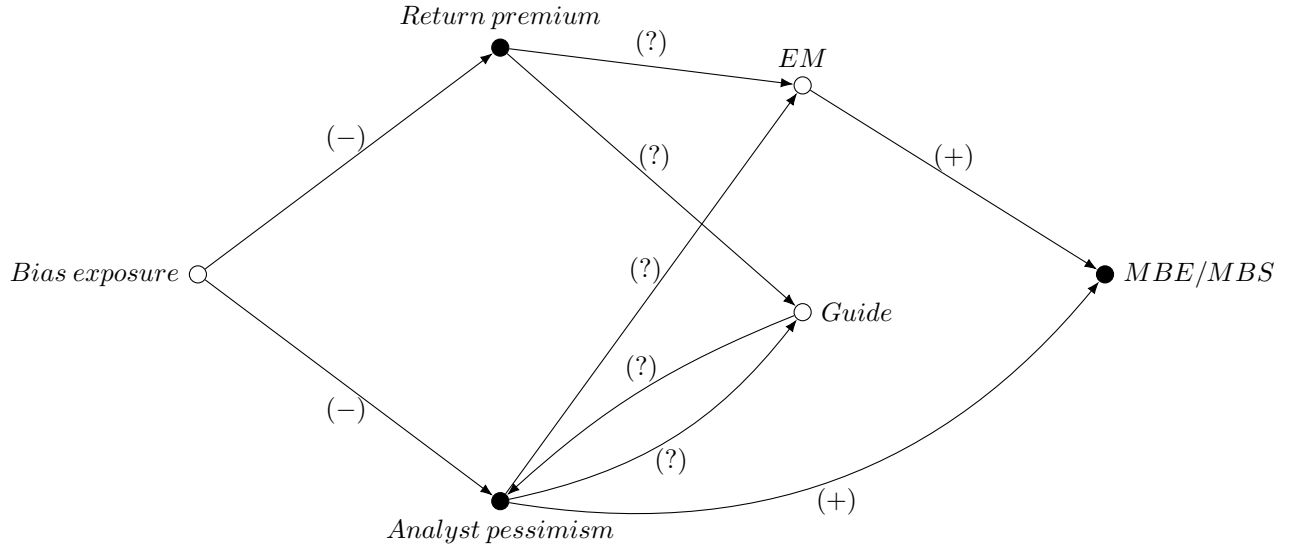


Figure 5

Plots of dynamic difference-in-differences coefficients in event-time

This figure presents the two-way fixed effects difference-in-differences estimations from Tables 5, 6, and 8 in event-time. To provide an understanding of the effects in a dynamic fashion and to assess the parallel trends assumption, the figure plots the coefficients obtained from regressions of the outcome variable on firm and year-quarter fixed effects and a set of indicator variables indicating the event year relative to the first year of Estimize coverage. The dots reflect the point estimates of the coefficients and the shaded areas capture the 95 percent confidence intervals based on robust standard errors adjusted for clustering by firm and year-quarter. The pre-coverage year (-1) is used as the baseline, while indicator variables pick up the difference-in-differences estimates for event years -3, -2, 0, +1, +2, and +3. Panel A presents the estimates using the incidence of meeting/beating both analyst earnings and revenue expectations as a dependent variable ($MBES_{Street}$), Panel B presents the estimates using announcement returns ($BHAR_{[0,1]}$) as dependent variable, Panel C presents the estimates using downward earnings guidance as a dependent variable ($Guide_{Down}^{EPS}$), and Panel D presents the estimates using downward revenue guidance as a dependent variable ($Guide_{Down}^{SAL}$).

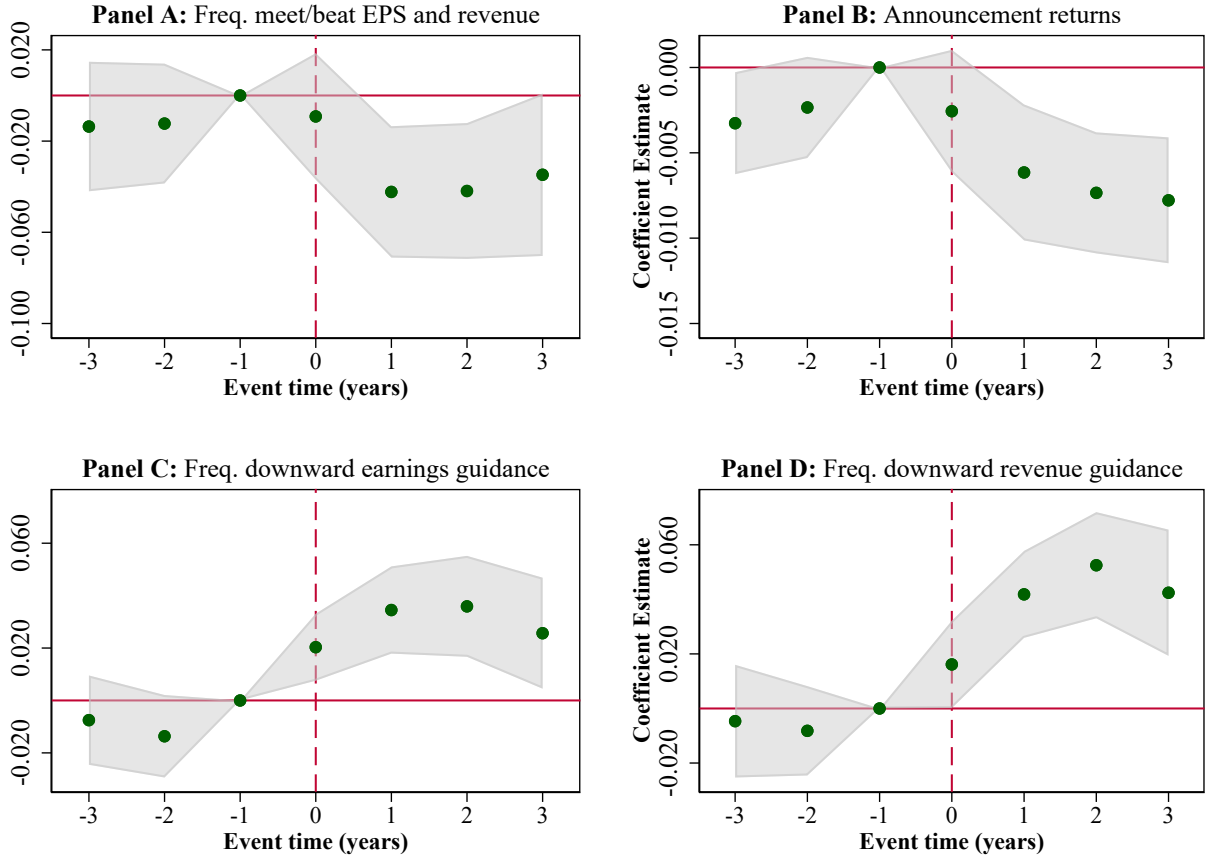


Table 1
Sample selection and descriptive statistics

Panel A: Sample selection details

	No. obs.	Firms
Unique firm-quarters in CRSP/Compustat for fiscal periods ending in calendar quarters 2012Q1-2020Q4	165,871	7,342
– Firms not listed on NYSE, AMEX, or NASDAQ	-103	-2
– No earnings surprise data in I/B/E/S	-41,349	-1,417
– Earnings announcement dates differ by > 2 trading days	-819	-14
– Earnings announcement delay > 120 days	-91	-2
– Earnings announcement return data missing from CRSP	-4,391	-312
– No forecasts on Estimize	-70,532	-2,927
Sample of Estimize firm-quarters	48,586	2,668
Extended sample including non-Estimize firms and firm-quarters (2010–2020)	128,182	5,456

Panel B: Sample and earnings forecast characteristics over time

Calendar year	Firm-quarters	Fraction I/B/E/S	Size decile	Crowdsourced		Sell-side	
				Median #forecasts consensus	Median forecast age	Median #forecasts consensus	Median forecast age
2012	1,614	0.128	8.9	6	6.2	14	69.8
2013	2,666	0.207	8.8	4	5.4	14	69.6
2014	3,933	0.291	8.7	5	16.8	13	69.1
2015	5,538	0.404	8.2	6	12.5	10	68.9
2016	6,959	0.518	7.9	7	18.5	9	70.0
2017	7,458	0.567	7.8	7	18.2	8	69.0
2018	7,182	0.544	7.8	7	19.7	8	68.7
2019	6,894	0.521	7.9	6	23.3	8	69.0
2020	6,342	0.474	8.1	5	21.9	9	58.8
Total	48,586	0.408	8.1	6	17.9	9	68.1

Panel C: Descriptive statistics for key variables in Estimize sample

	Mean	St.dev.	p10	Q1	Median	Q3	p90
ES_{Street} (\$)	0.046	0.181	-0.080	-0.010	0.030	0.090	0.200
ES_{Crowd} (\$)	0.015	0.199	-0.130	-0.030	0.010	0.070	0.180
MBE_{Street}	0.743	0.437	0	0	1	1	1
MBE_{Crowd}	0.620	0.485	0	0	1	1	1
SS_{Street} (%)	0.013	0.085	-0.048	-0.013	0.008	0.034	0.080
SS_{Crowd} (%)	-0.001	0.106	-0.067	-0.021	0.003	0.028	0.073
MBS_{Street}	0.612	0.487	0	0	1	1	1
MBS_{Crowd}	0.534	0.499	0	0	1	1	1
$BHAR_{[0,1]}$	0.000	0.094	-0.100	-0.044	0.000	0.045	0.100
$Size$	14,135	32,468	468	1,180	3,295	10,899	33,408
BTM	0.434	0.464	0.064	0.171	0.331	0.570	0.916
$BHAR_{[-5,-1]}$	0.001	0.043	-0.047	-0.020	0.001	0.023	0.049
$BHAR_{[-60,-6]}$	0.001	0.153	-0.174	-0.080	-0.002	0.075	0.171
N_{Street}	10.921	7.018	3.000	5.000	9.000	15.000	21.000
N_{Crowd}	12.769	18.489	2.000	4.000	6.000	14.000	29.000

Panel D: Descriptive statistics for key variables in extended sample

	Mean	St.dev.	p10	Q1	Median	Q3	p90
ES_{Street} (\$)	0.024	0.176	-0.100	-0.020	0.020	0.070	0.170
MBE_{Street}	0.670	0.470	0	0	1	1	1
SS_{Street} (%)	0.013	0.131	-0.076	-0.020	0.007	0.041	0.105
MBS_{Street}	0.585	0.493	0	0	1	1	1
$BHAR_{[0,1]}$	-0.001	0.088	-0.091	-0.038	0.000	0.037	0.088
$Size$	6,559	17,034	133	369	1,275	4,333	14,743
BTM	0.568	0.516	0.101	0.242	0.465	0.784	1.142
$BHAR_{[-5,-1]}$	0.001	0.045	-0.049	-0.022	0.000	0.022	0.050
$BHAR_{[-60,-6]}$	-0.001	0.164	-0.185	-0.086	-0.005	0.074	0.175
N_{Street}	7.768	6.291	2.000	3.000	6.000	11.000	17.000

Notes: Panel A presents the criteria used to construct the main sample of firm-quarters. The sample consists of firms in the intersection of CRSP, Compustat, I/B/E/S, and Estimize, with fiscal periods ending in calendar quarters 2012Q1 through 2020Q4. We follow [Dellavigna and Pollet \(2009\)](#) and identify the earnings announcement date from I/B/E/S and Compustat data, and [Johnson and So \(2018\)](#) by adjusting the announcement date to the next trading day when the announcement is made after trading hours. Panel B presents sample and earnings forecast characteristics over the calendar-years 2012–2020, displaying the number of firm-quarters in the final sample, the fraction of firm-quarters relative to the I/B/E/S universe, the average size of the sample firms based on CRSP size deciles, the median number and age of forecasts in the crowdsourced consensus earnings forecasts, and the median number and age of forecasts in the sell-side consensus earnings forecasts, respectively. Panel C presents descriptive statistics on the variables used in our analyses using the Estimize sample. Panel D presents descriptive statistics on the variables used in our analyses using the extended sample (2010–2020) including firms and firm-quarters not covered by Estimize. For presentation purposes, we winsorize all continuous variables (except the announcement return variable) at the 1st and 99th percentiles of their distributions. All variables are defined in Appendix B, except that variables ES_{Street} and ES_{Crowd} are presented in unscaled terms (\$), while SS_{Street} and SS_{Crowd} are presented as percentage deviations from the consensus forecast (%).

Table 2

Descriptive tests of market reactions to earnings and revenue surprises

Panel A: ERCs for analyst versus crowdsourced earnings and revenue surprises

	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$
ES_{Street}	0.051 (13.80)***	0.042 (11.96)***	0.019 (4.71)***	0.018 (3.74)***
ES_{Crowd}	0.043 (13.44)***	0.037 (12.11)***	0.027 (9.15)***	0.031 (8.22)***
SS_{Street}		0.031 (10.09)***	0.022 (7.03)***	0.022 (6.92)***
SS_{Crowd}		0.012 (3.97)***	0.009 (2.86)***	0.008 (2.45)**
$ES_{BiasAdj}$			0.039 (13.31)***	0.041 (13.57)***
$SS_{BiasAdj}$			0.015 (5.00)***	0.015 (4.58)***
Year-quarter FE	Included	Included	Included	Included
Require same actuals	-	-	-	Yes
n	48,586	48,586	48,006	40,099
Adj. R^2	0.096	0.113	0.122	0.127

Panel B: Premia for meeting analyst expectations when missing crowdsourced expectations

	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$
$MBE_{Street} (\beta_1)$	0.059 (25.07)***	0.016 (8.45)***	0.049 (22.28)***	0.013 (6.01)***
$MBE_{Street} \times Miss_{Crowd} (\beta_2)$	-0.032 (-19.03)***	-0.014 (-9.26)***	-0.026 (-17.33)***	-0.011 (-7.84)***
$MBS_{Street} (\beta_3)$			0.030 (20.59)***	0.013 (6.21)***
$MBS_{Street} \times Miss_{Crowd} (\beta_4)$			-0.015 (-9.15)***	-0.012 (-7.81)***
ES_{Street}		0.060 (14.13)***		0.063 (14.96)***
SS_{Street}		0.042 (18.12)***		0.026 (7.26)***
$\beta_1 + \beta_2$	0.027 [0.0000]	0.002 [0.1998]	0.023 [0.0000]	0.002 [0.4099]
$\beta_3 + \beta_4$			0.015 [0.0000]	0.001 [0.5078]
Year-quarter FE	Included	Included	Included	Included
Require same actuals	Yes	Yes	Yes	Yes
n	40,586	40,586	40,586	40,586
Adj. R^2	0.073	0.113	0.092	0.115

Panel C: Tests using a bias-adjusted Street earnings surprise as the benchmark

	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$
$MBE_{BiasAdj} (\beta_1)$	0.052 (28.64)***	0.005 (3.95)***	0.045 (25.81)***	0.003 (2.57)**
$MBE_{BiasAdj} \times Miss_{Crowd} (\beta_2)$	-0.027 (-11.59)***	-0.019 (-8.05)***	-0.023 (-9.64)***	-0.017 (-7.10)***
$MBS_{BiasAdj} (\beta_3)$			0.030 (19.35)***	0.015 (7.18)***
$MBS_{BiasAdj} \times Miss_{Crowd} (\beta_4)$			-0.016 (-8.52)***	-0.014 (-7.31)***
$ES_{BiasAdj}$		0.076 (19.05)***		0.077 (20.02)***
$SS_{BiasAdj}$		0.039 (16.48)***		0.022 (6.66)***
$\beta_1 + \beta_2$	0.025 [0.0000]	-0.014 [0.0000]	0.022 [0.0000]	-0.014 [0.0000]
$\beta_3 + \beta_4$			0.014 [0.0000]	0.001 [0.7488]
Year-quarter FE	Included	Included	Included	Included
Require same actuals	Yes	Yes	Yes	Yes
n	40,099	40,099	40,099	40,099
Adj. R^2	0.074	0.115	0.095	0.117

Notes: Panel A presents descriptive analyses of the market reactions associated with sell-side versus crowdsourced surprises based on forecasts of earnings and revenues. We estimate earnings response coefficients (ERCs) using OLS regressions of earnings announcement returns on measures of earnings and revenue surprise. Panel B presents tests of the incremental announcement returns associated with meeting/beating analyst expectations when firms simultaneously miss crowdsourced expectations. Given the regression setup, the coefficients on MBE_{Street} and MBS_{Street} capture the announcement return premia associated with meeting/beating both analyst and crowdsourced expectations. The coefficients on the interactions with $Miss_{Crowd}$ capture the difference in return premia associated with both meeting/beating analyst expectations and missing crowdsourced expectations. In Panel C, we perform the same analyses but replace the variables for meeting/beating analyst expectations with indicator variables that capture whether or not the firm meets/beats bias-adjusted forecasts of earnings and revenues ($MBE_{BiasAdj}$ and $MBS_{BiasAdj}$). Across the table panels, sample sizes vary due to the requirement of data availability for the bias-adjusted surprise variables and restricting the analyses to observations where the actuals earnings and revenue numbers of Estimote and I/B/E/S are similar (i.e., EPS actuals match to one decimal point and revenue actuals are within \$1 mln). All variables are defined in Appendix B. We rank the continuous independent variables into decile portfolios each quarter and transform the ranks to a $[-0.5, 0.5]$ scale. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. Calendar-year-quarter fixed effects are included but not tabulated. t -statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.

Table 3

Tests of the predictive value of crowdsourced forecasts for analyst forecast bias

Panel A: Meet/beat frequencies and announcement returns by bins of earnings forecast difference

	Earnings forecast difference (FD^{EPS}):						
	$\leq -3\text{¢}$	-2¢	-1¢	0¢	1¢	2¢	$\geq 3\text{¢}$
MBE_{Street}	0.587	0.646	0.666	0.734	0.791	0.798	0.751
$BHAR_{[0,1]}$	0.290	0.249	-0.007	0.137	0.098	0.062	-0.100

Panel B: Meet/beat frequencies and announcement returns by bins of revenue forecast difference

	Revenue forecast difference (FD^{SAL}):						
	$\leq -3\%$	-2%	-1%	0%	1%	2%	$\geq 3\%$
MBS_{Street}	0.483	0.547	0.555	0.596	0.662	0.675	0.648
$BHAR_{[0,1]}$	-0.003	-0.088	0.086	0.101	0.108	0.148	-0.220

Panel C: Regression tests of differences in meet/beat frequencies and announcement returns

	Earnings (FD^{EPS})		Revenue (FD^{SAL})	
	MBE_{Street}	$BHAR_{[0,1]}$	MBS_{Street}	$BHAR_{[0,1]}$
$\mathbb{1}(FD^{EPS} = 1\text{¢}) \text{ /// } \mathbb{1}(FD^{SAL} = 1\%)$	0.076 (8.05)***	-0.016 (-0.09)	0.054 (6.79)***	0.172 (1.24)
$\mathbb{1}(FD^{EPS} = 2\text{¢}) \text{ /// } \mathbb{1}(FD^{SAL} = 2\%)$	0.075 (8.64)***	-0.027 (-0.18)	0.064 (4.58)***	0.198 (0.93)
$\mathbb{1}(FD^{EPS} = 3\text{¢}) \text{ /// } \mathbb{1}(FD^{SAL} = 3\%)$	0.074 (6.57)***	-0.062 (-0.33)	0.044 (3.28)***	-0.237 (-0.93)
<i>Size</i>	0.068 (4.72)***	-0.358 (-0.97)	-0.000 (-0.00)	-0.313 (-0.84)
<i>BTM</i>	-0.020 (-2.13)**	0.292 (1.46)	-0.062 (-4.27)***	0.200 (0.87)
$BHAR_{[-5,-1]}$	0.047 (5.02)***	-0.268 (-1.42)	0.085 (6.72)***	-0.335 (-1.42)
$BHAR_{[-60,-6]}$	0.089 (8.85)***	0.336 (1.27)	0.138 (10.13)***	0.205 (0.73)
N_{Street}	0.004 (0.30)	0.337 (1.19)	-0.002 (-0.17)	0.419 (1.48)
N_{Crowd}	-0.003 (-0.31)	-0.585 (-2.65)**	0.025 (2.22)**	-0.947 (-3.38)***
$MBE_{12q} \text{ /// } MBS_{12q}$	0.510 (28.78)***	0.427 (1.22)	0.475 (19.39)***	0.043 (0.12)
Year-quarter FE	Included	Included	Included	Included
n	26,579	26,579	24,361	24,361
Adj. R^2	0.098	0.003	0.111	0.002

Notes: This table presents tests of the predictive signals in crowdsourced forecasts for the bias in analyst forecasts and the market pricing of these signals. Panel A presents the frequencies of meeting/beating analyst earnings forecasts and earnings announcement returns for bins of the pre-earnings announcement difference between the crowdsourced and analyst consensus forecasts of earnings, in cents per share. A positive (negative) difference implies the crowdsourced consensus is higher (lower) than the analyst consensus forecast. Panel B presents similar statistics for differences in forecasts of revenues, where the differences are denoted in (rounded) percentage differences from the analyst consensus forecast. Panel C presents results from OLS regressions of either the meet/beat indicator or announcement returns on indicator variables for whether the difference in earnings (revenue) forecasts is equal to 1, 2, or 3 cents per share (percent) and control variables. The samples consist of observations with a difference in earnings (revenue) forecasts with 3 cents per share (3 percent), excluding observations with no difference. We rank the continuous independent variables into decile portfolios each quarter and transform the ranks to a $[-0.5, 0.5]$ scale. All variables are defined in Appendix B. For presentation purposes, we multiply the values of the announcement return by 100. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. Calendar-year-quarter fixed effects are included but not tabulated. t -statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.

Table 4

Mispricing of forecast bias before and after Estimimize coverage

Panel A: Portfolios formed on measures of prior pessimistic bias in earnings and sales forecasts

	Avg.	Mean by $PessEPS$ portfolio					Q5-Q1	t -stat
		Q1	Q2	Q3	Q4	Q5		
MBE_{Street}	0.667	0.479	0.583	0.662	0.749	0.865	0.387	59.01***
$BHAR_{[0,1]}$	-0.037	-0.410	-0.027	0.032	0.056	0.165	0.575	5.51***
$BHAR_{[0,1]}$ pre-Estimize	0.319	-0.064	0.277	0.329	0.313	0.618	0.682	5.87***
$BHAR_{[0,1]}$ post-Estimize	0.062	-0.103	0.165	0.157	-0.002	0.057	0.160	1.16

	Avg.	Mean by $PessSAL$ portfolio					Q5-Q1	t -stat
		Q1	Q2	Q3	Q4	Q5		
MBS_{Street}	0.583	0.408	0.513	0.575	0.648	0.769	0.362	36.89***
$BHAR_{[0,1]}$	-0.028	-0.394	-0.086	0.061	0.186	0.091	0.485	5.12***
$BHAR_{[0,1]}$ pre-Estimize	0.326	0.060	-0.027	0.433	0.631	0.503	0.443	2.48**
$BHAR_{[0,1]}$ post-Estimize	0.065	-0.113	0.127	0.061	0.145	0.082	0.195	1.44

Panel B: Regression analyses of mispricing before and after Estimimize coverage

	$Pess$ variable measured as:		
	$PessEPS$ $BHAR_{[0,1]}$	$PessSAL$ $BHAR_{[0,1]}$	$PessEPS+SAL$ $BHAR_{[0,1]}$
<u>Test variables:</u>			
$Pess$ (β_1)	0.681 (3.63)***	0.769 (5.05)***	0.942 (5.59)***
$Pess \times Estimimize$ (β_2)	0.131 (0.58)	0.125 (0.58)	0.023 (0.11)
$Pess \times Post$ (β_3)	-0.651 (-3.45)***	-0.527 (-2.46)**	-0.621 (-3.15)***
<u>Control variables:</u>			
$Estimimize$	0.870 (8.14)***	0.985 (9.30)***	0.922 (8.71)***
$Post$	-0.469 (-3.90)***	-0.491 (-4.02)***	-0.498 (-4.03)***
$Size$	0.186 (0.84)	0.189 (0.84)	0.131 (0.59)
BTM	0.875 (6.02)***	0.896 (6.17)***	0.898 (6.18)***
$BHAR_{[-5,-1]}$	-0.667 (-5.65)***	-0.709 (-6.12)***	-0.717 (-6.18)***
$BHAR_{[-60,-6]}$	0.298 (2.30)**	0.377 (2.84)***	0.371 (2.82)***
N_{Street}	-0.258 (-1.39)	-0.203 (-1.13)	-0.228 (-1.27)
$\beta_1 + \beta_2 + \beta_3$	0.161 [0.3004]	0.367 [0.0088]	0.344 [0.0299]
Year-quarter FE	Included	Included	Included
n	123,924	118,834	118,834
Adjusted R^2	0.004	0.004	0.004

Notes: This table presents mispricing tests for an extended sample of firm-quarters from 2010–2020. All variables are defined in Appendix B. Panel A presents average meet/beat frequencies and announcement returns for quintiles portfolios formed on ex-ante forecast pessimism measures ($Pess_{EPS}$ and $Pess_{SAL}$). Panel B presents regression tests that compare the relation between the pessimism measures and announcement returns for Estimate firms before and after coverage, and relative to non-Estimate covered firms. A significant positive difference (coefficient) in Panel A (Panel B) provides evidence consistent with mispricing. We rank the independent variables into decile portfolios each quarter and transform the ranks to a $[-0.5, 0.5]$ scale. For presentation purposes, we multiply the values of the announcement return by 100. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. Calendar-year-quarter fixed effects are included but not tabulated. t -statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.

Table 5

Overall effects on meet/beat frequency after Estimote coverage

Panel A: Generalized difference-in-differences estimation

	MBE_{Street}	$MJBE_{Street}$	MBS_{Street}	$MJBS_{Street}$	$MBES_{Street}$
<i>Post</i>	-0.015 (-1.75)*	-0.034 (-3.14)***	-0.031 (-3.33)***	-0.023 (-3.21)***	-0.032 (-3.40)***
Firm FE	Included	Included	Included	Included	Included
Year-quarter FE	Included	Included	Included	Included	Included
n	128,182	63,320	128,182	67,748	128,182
Adj. R^2	0.131	0.203	0.118	0.130	0.127

Panel B: Generalized difference-in-differences estimation including control variables

	MBE_{Street}	$MJBE_{Street}$	MBS_{Street}	$MJBS_{Street}$	$MBES_{Street}$
<i>Post</i>	-0.012 (-1.44)	-0.031 (-2.93)***	-0.026 (-2.82)***	-0.021 (-2.90)***	-0.028 (-3.00)***
<u>Control variables:</u>					
<i>Size</i>	0.034 (1.54)	0.070 (2.83)***	0.071 (2.08)**	0.090 (3.83)***	0.059 (2.06)**
<i>BTM</i>	-0.014 (-1.23)	-0.032 (-2.41)**	-0.040 (-2.87)***	-0.024 (-1.98)*	-0.047 (-3.46)***
<i>BHAR</i> _[-5,-1]	0.054 (13.12)***	0.029 (5.67)***	0.063 (12.62)***	0.034 (7.25)***	0.067 (13.68)***
<i>BHAR</i> _[-60,-6]	0.077 (15.52)***	0.048 (8.89)***	0.102 (16.60)***	0.047 (9.19)***	0.103 (16.14)***
<i>N_{Street}</i>	-0.019 (-1.51)	-0.020 (-1.29)	-0.064 (-5.57)***	-0.000 (-0.04)	-0.041 (-3.62)***
Firm FE	Included	Included	Included	Included	Included
Year-quarter FE	Included	Included	Included	Included	Included
n	128,182	63,320	128,182	67,748	128,182
Adj. R^2	0.135	0.205	0.125	0.132	0.134

Notes: This table presents results from OLS regressions (linear probability models) of meet/beat indicator variables on firm fixed effects, year-quarter fixed effects, and indicator variable *Post*. *Post* equals 0 for firm-quarters before Estimote coverage and control firms never covered on Estimote, and equals 1 for firm-quarters after coverage on Estimote. The coefficient on *Post* captures the difference-in-differences in meet/beat frequency around firms' addition to Estimote relative to all firm-quarters not (yet) covered on Estimote. Treatment firms are those added to Estimote in any quarter during 2012–2020 and control firms are those never added to Estimote during our sample period 2012–2020. Panel A presents results without control variables and Panel B presents results including control variables. All variables are defined in Appendix B. We rank the independent variables in Panel B into decile portfolios each quarter and transform the ranks to a $[-0.5, 0.5]$ scale. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. Firm and calendar-year-quarter fixed effects are included but not tabulated. *t*-statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.

Table 6

Average earnings announcement returns before and after Estimize coverage

Panel A: Generalized difference-in-differences estimation

	$BHAR_{[0,1]}$	MBE_{Street}		MBS_{Street}	
		0	1	0	1
		$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$
<i>Post</i>	-0.515 (-4.18)***	-0.405 (-2.07)**	-0.436 (-3.32)***	-0.176 (-1.00)	-0.628 (-4.46)***
Firm FE	Included	Included	Included	Included	Included
Year-quarter FE	Included	Included	Included	Included	Included
n	128,182	42,236	85,946	53,346	74,836
Adj. R^2	0.025	0.086	0.047	0.074	0.043

Panel B: Generalized difference-in-differences estimation including control variables

	$BHAR_{[0,1]}$	MBE_{Street}		MBS_{Street}	
		0	1	0	1
		$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$	$BHAR_{[0,1]}$
<i>Post</i>	-0.474 (-3.83)***	-0.442 (-2.29)**	-0.463 (-3.28)***	-0.235 (-1.50)	-0.619 (-4.07)***
<u>Control variables:</u>					
ES_{Street}	7.391 (39.49)***	11.335 (14.62)***	7.685 (31.22)***	6.944 (31.04)***	7.431 (37.11)***
SS_{Street}	4.159 (25.70)***	3.293 (17.36)***	4.180 (21.49)***	6.627 (15.78)***	4.791 (16.15)***
<i>Size</i>	-4.271 (-8.23)***	-4.914 (-6.32)***	-4.777 (-8.40)***	-5.070 (-7.19)***	-4.275 (-6.60)***
<i>BTM</i>	1.512 (6.01)***	2.465 (6.10)***	1.473 (4.49)***	2.500 (8.14)***	1.108 (3.38)***
$BHAR_{[-5,-1]}$	-1.222 (-10.65)***	-1.136 (-6.38)***	-1.418 (-10.55)***	-1.464 (-11.15)***	-1.137 (-8.12)***
$BHAR_{[-60,-6]}$	-0.381 (-3.05)***	-0.733 (-4.61)***	-0.465 (-3.02)***	-0.777 (-5.25)***	-0.525 (-3.60)***
N_{Street}	-0.129 (-0.54)	-0.545 (-1.53)	0.301 (1.10)	-0.321 (-0.98)	-0.006 (-0.02)
Firm FE	Included	Included	Included	Included	Included
Year-quarter FE	Included	Included	Included	Included	Included
n	128,182	42,236	85,946	53,346	74,836
Adj. R^2	0.136	0.115	0.115	0.146	0.127

Notes: This table presents results from OLS regressions of earnings announcement returns on firm fixed effects, year-quarter fixed effects, and indicator variable *Post*. *Post* equals 0 for firm-quarters before Estimize coverage and control firms never covered on Estimize, and equals 1 for firm-quarters after coverage on Estimize. The coefficient on *Post* captures the difference-in-differences in announcement returns around firms' addition to Estimize relative to all firm-quarters not (yet) covered on Estimize. Treatment firms are those added to Estimize in any quarter during 2012–2020 and control firms are those never added to Estimize during our sample period 2012–2020. Panel A presents results without control variables and Panel B presents results including control variables. All variables are defined in Appendix B. We rank the independent variables in Panel B into decile portfolios each quarter and transform the ranks to a $[-0.5, 0.5]$ scale. For presentation purposes, we multiply the values of the announcement return by 100. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. Firm and calendar-year-quarter fixed effects are included but not tabulated. *t*-statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.

Table 7

Tests of abnormal accruals associated with meeting expectations

Panel A: Abnormal accruals associated with meeting Street vs. crowdsourced expectations

	Jones <i>WCA</i>	Jones <i>WCA</i>	Jones <i>WCA</i>	Mod-Jones <i>WCA</i>	Mod-Jones <i>WCA</i>	Mod-Jones <i>WCA</i>
MBE_{Street}	0.130 (3.26)***		0.083 (2.26)**	0.160 (4.15)***		0.059 (1.60)
MBE_{Crowd}		0.111 (3.24)***			0.170 (5.11)***	
MBE_{Both}			0.060 (2.09)**			0.129 (4.80)***
Determinants	Included	Included	Included	Included	Included	Included
n	35,007	35,007	35,007	35,007	35,007	35,007
Adj. R^2	0.347	0.347	0.347	0.378	0.378	0.378

Panel B: Abnormal accruals in quarters after firms miss crowdsourced expectations

	Jones <i>WCA</i>	Jones <i>WCA</i>	Mod-Jones <i>WCA</i>	Mod-Jones <i>WCA</i>
$MBE_{Street,t-1}$	0.000 (0.01)	-0.003 (-0.05)	-0.062 (-1.33)	-0.071 (-1.25)
$MBE_{Street,t-1} \times Miss_{Crowd,t-1}$		0.093 (1.96)*		0.142 (3.08)***
Determinants	Included	Included	Included	Included
n	34,793	31,233	34,793	31,233
Adj. R^2	0.347	0.346	0.378	0.375

Panel C: Abnormal accruals associated with meeting expectations after Estimize coverage

	Jones <i>WCA</i>	Mod-Jones <i>WCA</i>
MBE_{Street}	0.294 (7.18)***	0.380 (8.67)***
$MBE_{Street} \times Estimze$	-0.166 (-2.53)**	-0.203 (-2.91)***
$MBE_{Street} \times Post$	-0.049 (-0.70)	-0.062 (-0.88)
<i>Estimze</i>	0.058 (0.97)	0.061 (0.95)
<i>Post</i>	-0.083 (-1.29)	-0.104 (-1.43)
Determinants	Included	Included
n	84,579	84,579
Adj. R^2	0.275	0.295

Notes: This table presents tests of accrual-based earnings management associated with meeting/beating analyst versus crowdsourced earnings forecasts in Panels A and B. In Panel C, we examine the effects of the addition of firms to Estimize on the use of accrual-based earnings management to meet/beat expectations. The untabulated determinants include the accrual model variables from equation (1) that are interacted with two-digit SIC industry-year indicator variables. All variables are defined in Appendix B. For presentation purposes, we multiply the values of working capital accruals (*WCA*) by 100. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. *t*-statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.

Table 8
Guidance frequency after Estimize coverage

Panel A: Generalized difference-in-differences estimation

	$Guide^{EPS}$	$Guide_{Down}^{EPS}$	$Guide_{Up}^{EPS}$	$Guide^{SAL}$	$Guide_{Down}^{SAL}$	$Guide_{Up}^{SAL}$
<i>Post</i>	0.020 (3.04)***	0.028 (4.44)***	-0.003 (-0.90)	0.029 (4.35)***	0.038 (6.20)***	0.003 (1.08)
Firm FE	Included	Included	Included	Included	Included	Included
Year-quarter FE	Included	Included	Included	Included	Included	Included
n	128,182	128,182	128,182	128,182	128,182	128,182
Adj. R^2	0.734	0.425	0.325	0.682	0.351	0.328

Panel B: Generalized difference-in-differences estimation including control variables

	$Guide^{EPS}$	$Guide_{Down}^{EPS}$	$Guide_{Up}^{EPS}$	$Guide^{SAL}$	$Guide_{Down}^{SAL}$	$Guide_{Up}^{SAL}$
<i>Post</i>	0.018 (2.74)***	0.026 (4.08)***	-0.003 (-0.88)	0.027 (3.97)***	0.034 (5.58)***	0.003 (1.07)
<u>Control variables:</u>						
<i>Size</i>	0.131 (6.61)***	0.114 (7.74)***	0.025 (2.64)**	0.091 (4.39)***	0.057 (3.38)***	0.029 (2.92)***
<i>BTM</i>	0.002 (0.18)	0.013 (1.47)	-0.004 (-0.93)	0.000 (0.01)	0.027 (2.84)***	-0.012 (-2.46)**
<i>BHAR</i> _[-5,-1]	0.003 (1.41)	-0.019 (-6.88)***	0.008 (5.33)***	0.000 (0.16)	-0.029 (-8.18)***	0.010 (5.19)***
<i>BHAR</i> _[-60,-6]	-0.005 (-1.82)*	-0.035 (-10.74)***	0.010 (5.73)***	-0.005 (-1.63)	-0.047 (-11.27)***	0.012 (5.95)***
<i>N_{Street}</i>	0.076 (7.35)***	0.068 (7.61)***	0.010 (2.15)**	0.100 (8.74)***	0.084 (8.32)***	0.024 (4.21)***
Firm FE	Included	Included	Included	Included	Included	Included
Year-quarter FE	Included	Included	Included	Included	Included	Included
n	128,182	128,182	128,182	128,182	128,182	128,182
Adj. R^2	0.735	0.428	0.326	0.684	0.355	0.329

Notes: This table presents results from OLS regressions (linear probability models) of guidance indicator variables on firm fixed effects, year-quarter fixed effects, and indicator variable *Post*. *Post* equals 0 for firm-quarters before Estimize coverage and control firms never covered on Estimize, and equals 1 for firm-quarters after coverage on Estimize. The coefficient on *Post* captures the difference-in-differences in guidance frequency around firms' addition to Estimize relative to all firm-quarters not (yet) covered on Estimize. Treatment firms are those added to Estimize in any quarter during 2012–2020 and control firms are those never added to Estimize during our sample period 2012–2020. Panel A presents results without control variables and Panel B presents results including control variables. All variables are defined in Appendix B. We rank the independent variables in Panel B into decile portfolios each quarter and transform the ranks to a $[-0.5, 0.5]$ scale. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. Firm and calendar-year-quarter fixed effects are included but not tabulated. *t*-statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.

Table 9
Stacked difference-in-differences regression results

Panel A: Stacked regressions without controls

	$MBES_{Street}$	$BHAR_{[0,1]}$	$Guide_{Down}^{EPS}$	$Guide_{Down}^{SAL}$
<i>Post</i>	-0.033 (-3.15)***	-0.553 (-4.30)***	0.030 (4.16)***	0.036 (5.46)***
Cohort-firm FE	Included	Included	Included	Included
Cohort-year-quarter FE	Included	Included	Included	Included
n	431,382	431,382	431,382	431,382
Adj. R^2	0.134	0.050	0.505	0.444

Panel B: Stacked regressions with controls

	$MBES_{Street}$	$BHAR_{[0,1]}$	$Guide_{Down}^{EPS}$	$Guide_{Down}^{SAL}$
<i>Post</i>	-0.030 (-2.97)***	-0.603 (-4.67)***	0.029 (4.05)***	0.033 (5.17)***
<u>Control variables:</u>				
<i>Size</i>	-0.080 (-2.22)**	-8.977 (-7.20)***	0.074 (4.63)***	0.023 (1.06)
<i>BTM</i>	0.033 (1.41)	4.163 (8.32)***	-0.002 (-0.20)	0.006 (0.38)
$BHAR_{[-5,-1]}$	0.056 (8.34)***	-1.036 (-6.77)***	-0.010 (-3.49)***	-0.022 (-6.21)***
$BHAR_{[-60,-6]}$	0.071 (8.43)***	0.375 (2.71)**	-0.019 (-5.22)***	-0.024 (-5.80)***
N_{Street}	-0.054 (-2.87)***	-1.083 (-2.63)**	0.060 (6.54)***	0.078 (8.44)***
Cohort-firm FE	Included	Included	Included	Included
Cohort-year-quarter FE	Included	Included	Included	Included
n	431,382	431,382	431,382	431,382
Adj. R^2	0.138	0.063	0.507	0.446

Notes: This table presents results from stacked event-by-event OLS regressions (linear probability models in the second through fourth columns). We treat each of the 20 calendar quarters during 2012–2016 in which firms are added to Estimote as a cohort and create 20 separate datasets. The treatment firms in a cohort are those added to Estimote in that quarter and control firms are those not included on Estimote during the event window from 8 quarters before to 8 quarters after the cohort-quarter. We create a stacked dataset by pooling the 20 event-specific “clean 2×2 ” datasets and aligning them in event-time. For each of the datasets, we create separate sets of fixed effects such that the firm and year-quarter fixed effects are now cohort-firm and cohort-year-quarter fixed effects. *Post* equals 1 in the eight quarters following initial coverage for firms covered by Estimote, and equals 0 for the eight quarters before initial coverage these firms and all firm-quarters of firms not covered by Estimote during the event window. Panel A presents results without control variables and Panel B presents results including control variables. For presentation purposes, we multiply the values of the announcement return by 100. All variables are defined in Appendix B. We rank the independent variables in Panel B into decile portfolios each quarter and transform the ranks to a $[-0.5, 0.5]$ scale. Robust standard errors are adjusted for clustering by firm and calendar-year-quarter. *t*-statistics are presented in parentheses below the coefficient estimates and ***, **, and * reflect statistical significance at the levels of 0.01, 0.05, and 0.10, respectively.